QUIVER VARIETIES, SLODOWY SLICES AND AFFINE GRASSMANNIANS

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1. Introduction

In this paper, we want to find the isomorphism of quantization of affine Grassmannian slices, quiver varieties and nilpotent Slodowy slices beyond type A.

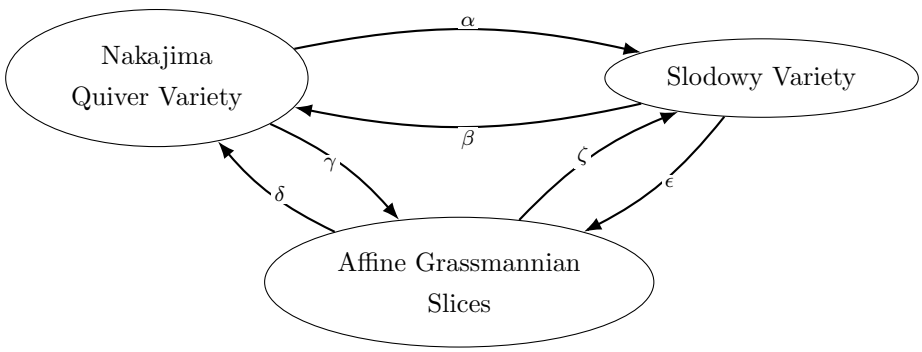


Figure 1. Relations among Nakajima quiver varieties, Slodowy varieties and affine Grassmannian slices.

1.1. **Histories.** In the part of Slodowy slices and quiver varieties, Losev [Los12] has used Maffei's result[Maf00] to prove that the quantum Hamiltonian reductions associated to quiver varieties of type A are isomorphic to parabolic W-algebras of type A. [HL14] hope that their results on quiver varieties of type D will have a similar application. Consider a quiver variety $\mathfrak{M}^{A_{2n-1}}(\mathbf{v},\mathbf{w})$ of type A_{2n-1}, for a positive integer n, where the data $\mathbf{v}=(v_1,\cdots,v_{2n-1})$, $\mathbf{w}=(w_1,\cdots,w_{2n-1})$ are symmetric under the Dynkin diagram involution, in the sense that $v_i=v_{2n-i}$ and $w_i=w_{2n-i}$ for all i. Note that the condition $w_i=w_{2n-i}$ means that the quiver variety corresponds, in Nakajima's theory, to a weight space of a self-dual representation of $\mathfrak{sl}_{2n}$. If we assume that $\mathbf{w}=(0,\cdots,0,1,0\cdots,0,1,0,\cdots,0)$, where the ones are in positions k and $2n-k$ for some $1\leq k\leq n$, the $k=n$ case giving $\mathbf{w}=(0,\cdots,0,2,0,\cdots,0)$. Thus, the associated representation of $\mathfrak{sl}_{2n}$, namely Maffei's result [Maf00] says that $\mathfrak{M}^{A_{2n-1}}(v,w)$ is isomorphic to a certain resolution of a closed subvariety of the Slodowy slice to the orbit with Jordan type $(2n-k,k)$ in the nilpotent cone of $\mathfrak{sl}_{2n}$ (a two-row Slodowy slice).

However, Maffei's isomorphism is given by a recursive characterization rather than by an explicit formula. In the works of Mirković and Vybornov [MV03] they construct Nakajima's quiver varieties of type A in terms of affine Grassmannians of type A, giving a compactification of quiver varieties and a decomposition of affine Grassmannians into a disjoint union of quiver varieties. Consequently, singularities of quiver varieties, nilpotent orbits and affine Grassmannians are the same in type A. In[MV08], In type A the two author relate Nakajima's quiver varieties, conjugacy classes of matrices, and Beilinson-Drinfeld Grassmannians, which embeds quiver varieties into Beilinson-Drinfeld Grassmannians. From the point of view of Nakajima's quiver varieties the construction provides a compactification of quiver varieties. From the point of view of nilpotent orbits they construct new transverse slices to nilpotent orbits naturally arising from affine Grassmannians. From the point of view of affine Grassmannians a decomposition of an affine Grassmannian into a disjoint union of quiver varieties is given. What they did here is close to Maffei's work, and could be considered as a classical analog of the Brundan-Kleshchev presentation[BK06]. In [HL14], the authors also suggest that Maffei's result that quiver varieties of type A are isomorphic to Slodowy varieties of type A is supplemented by a result that should be thought of as dual to it: namely, that resolutions of slices in the affne Grassmannian of type A are also isomorphic to Slodowy varieties of type A. The (partly conjectural) duality here is that of symplectic dual pairs, in the sense of [Bra+10][Remark 1.5], . However, while Maffei uses Slodowy's normal slices to nilpotent orbits, Mirković–Vybornov use different slices suggested by the relation to affine Grassmannians, and this makes the construction explicit, while Maffei's approach is based on an existence result.

Inspired by Maffei's work, Henderson and Licata [HL14] Showed that the fixed-point subvariety of a quiver variety under a diagram automorphism is a disconnected union of quiver varieties for the split-quotient quiver. further, they obtained the following result. Let $\widetilde{S}^{C_n}_{(2n-k,k)}$ and $\widetilde{S}^{D_n}_{(2n-k,k)}$ denote the Slodowy slices to the orbits with Jordan type $(2n-k,k)$ in the nilpotent cones of sp_{2n} and so_{2n}, respectively, the former being defined when k is even or $k=n$, and the latter being defined when k is odd or $k=n$. Let

$\widetilde{S}^{C_n}_{(2n-k,k)}$ and $\widetilde{S}^{D_n}_{(2n-k,k)}$ denote the Springer resolutions of these slices. We have isomorphisms of varieties

$$\widetilde{\mathcal{S}}^{\mathrm{C}_n}_{(2n-k,k)} \cong \begin{cases} \mathfrak{M}^{\mathrm{D}_{n+1}}((1,2,\cdots,k,k,\cdots,k,\frac{k}{2},\frac{k}{2}),(0,\cdots,0,1,0,\cdots,0)) \\ \text{ if } k<n(k \text{ even }), \text{ where the 1 on the right is in position } k, \\ \mathfrak{M}^{\mathrm{D}_{n+1}}((1,2,\cdots,n-1,\frac{n}{2},\frac{n}{2}),(0,\cdots,0,1,1)) \\ \text{ if } k=n \text{ is even}, \\ \mathfrak{M}^{\mathrm{D}_{n+1}}((1,2,\cdots,n-1,\frac{n-1}{2},\frac{n+1}{2}),(0,\cdots,0,2)) \\ \text{ if } k=n \text{ is odd}, \end{cases}$$

$$\widetilde{\mathcal{S}}^{\mathrm{D}_n}_{(2n-k,k)} \cong \begin{cases} \mathfrak{M}^{\mathrm{D}_{n+1}}((1,2,\cdots,k,k,\cdots,k,\frac{k-1}{2},\frac{k+1}{2}),(0,\cdots,0,1,0,\cdots,0)) \\ \text{ if } k<n(k \text{ odd }), \text{ where the 1 on the right is in position } k, \\ \mathfrak{M}^{\mathrm{D}_{n+1}}((1,2,\cdots,n-1,\frac{n-1}{2},\frac{n+1}{2}),(0,\cdots,0,1,1)) \\ \text{ if } k=n \text{ is odd}, \\ \mathfrak{M}^{\mathrm{D}_{n+1}}((1,2,\cdots,n-1,\frac{n}{2},\frac{n}{2}),(0,\cdots,0,2)) \\ \text{ if } k=n \text{ is even}. \end{cases}$$

In each case they also proved two further isomorphisms: between the Slodowy slice $\mathcal{S}^{\mathrm{C}_n}_{(2n-k,k)}$ or $\mathcal{S}^{\mathrm{D}_n}_{(2n-k,k)}$ and the affine variety $\mathfrak{M}^{\mathrm{D}_{n+1}}_1(\cdot,\cdot)$; and between the Springer fibre at the base-point of the slice, i.e. $\mathcal{B}^{\mathrm{C}_n}_{(2n-k,k)}$ or $\mathcal{B}^{\mathrm{D}_n}_{(2n-k,k)}$ as appropriate, and the projective variety $\mathfrak{L}^{\mathrm{D}}_{n+1}(\cdot,\cdot)$. Moreover, the action on $\widetilde{\mathcal{S}}^{\mathrm{C}_n}_{(2n-k,k)}$ or $\widetilde{\mathcal{S}}^{\mathrm{D}_n}_{(2n-k,k)}$ of the stabilizer of the $\mathfrak{sl}_2$-triple corresponds in each case to a natural reductive group action on the right-hand side. What's more, [Li18] proved that

$$\begin{tikzcd} \widetilde{\mathcal{S}}^{\mathfrak{sp}\,\widetilde{\mathrm{w}}}_{\mu',\lambda} \arrow[r, "\cong"] \arrow[d, "\Pi_{\widetilde{\mathrm{v}},\widetilde{\mathrm{w}}}"'] & \widetilde{\mathcal{S}}^{\mathfrak{o}\,\widetilde{\mathrm{w}}}_{\widetilde{\mu}',\widetilde{\lambda}} \arrow[d, "\Pi_{\widetilde{\mathrm{v}},\widetilde{\mathrm{w}}}"] \\ \mathcal{S}^{\mathfrak{sp}\,\widetilde{\mathrm{w}}}_{\mu',\lambda} \arrow[r, "\cong"] & \mathcal{S}^{\mathfrak{o}\,\widetilde{\mathrm{w}}}_{\widetilde{\mu}',\widetilde{\lambda}} \end{tikzcd}$$

which implies the following isomorphism of the Springer resolutions of the nilpotent Slodowy slices, which proves a conjecture in [[HL14],1.3].

$$\widetilde{\mathcal{S}}^{\mathfrak{sp}_n}_{n^1,k^1(n-k)^1} \cong \widetilde{\mathcal{S}}^{\mathfrak{o}_{n+2}}_{1^1(n+1)^1,(k+1)^1(n+1-k)^1}.$$

These works generalised Maffei's result in type A.

Losev used Maffei's result to work on type A quantized cases[Los12]. he showed that quantizations of Kleinian singularities obtained by three different constructions are isomorphic to each other. The constructions are via symplectic reflection algebras, quantum Hamiltonian reduction, and W-algebras. Next, he proved that parabolic W-algebras in type A are isomorphic to quantum Hamiltonian reductions associated with quivers of type A. But the proof of Losev did not give us the isomorphism explicitly.

1.2. **Quantization of Slodowy Slices.** Let $\mathfrak{g}$ be a simple complex Lie algebra and consider the quotient map $\varphi : \mathfrak{g} \to \mathfrak{g}//G$ for the action of the adjoint group G of $\mathfrak{g}$. The nullfibre $N = \varphi^{-1}(0)$ consists of the nilpotent elements in $\mathfrak{g}$ and is called the nilpotent cone. It is an irreducible variety and decomposes into finitely many orbits. The dense orbit is called the regular orbit and denoted by $\mathcal{O}_{\mathrm{reg}}$, its complement $N \setminus \mathcal{O}_{\mathrm{reg}}$ is again irreducible, its dense orbit is called the subregular orbit and denoted by $\mathcal{O}_{\mathrm{sub}}$.

A slice to the adjoint orbit of a nilpotent element $e \in N$ can be constructed as follows. The Jacobson–Morozov theorem allows one to find elements $h, y \in \mathfrak{g}$ such that e, h and y form an sl_2-triple, i.e. satisfy the standard commutator relations $[h, e] = 2x$, $[h, y] = -2y$ and $[e, y] = h$. The affine space

$$S = e + \mathrm{Ker}(\mathrm{ad}\, y)$$

is a special transversal slice [Slo80] to the orbit through e. We follow the tradition to refer to this choice as a Slodowy slice. The nilpotent part of a Slodowy slice is not an isolated surface singularity, and so, there is no versal theory for deformations. It turns out that the correct generalisation is given by realising the Slodowy slice as a Poisson variety via Hamiltonian reduction, following [GG02]The main result of [LNS12] gives a necessary and sufficient condition for the Slodowy slice to be a universal Poisson deformation of its nilpotent part.

Qantization of Slodowy slices are studied using W-algebras and Yangians. Let X be a conic symplectic singularity. Namikawa has shown in [Nam06; Nam10]that the functor PD_X of Poisson deformations is representable, and Losev has shown in [Los12] that the universal Poisson deformation can be quantized. What is more, Losev's quantization admits a very nice universal property. In [Amb+23] the functor Q_X of quantizations of flat Poisson

deformations of X was introduced, and it was explained that the universal property is equivalent to the representability of the functor of quantizations. The results of Namikawa and Losev were upgraded in [AT23] to the equivariant setting, with respect to a group of graded Poisson automorphisms of the underlying singularity. they gave a complete description of the cases in which the finite W-algebra is a universal filtered quantization of the slice, building on the work of Lehn–Namikawa–Sorger. This leads to a near-complete classification of the filtered quantizations of nilpotent Slodowy slices.

The work[BK06] of Brundan and Kleshchev, where shifted Yangians were found, gives a presentation for the finite W-algebra associated to a nilpotent matrix in the general linear Lie algebra over $\mathbb{C}$. In the special case that the nilpotent matrix consists of n Jordan blocks each of the same size l, the presentation is that of the Yangian of level l associated to $\mathfrak{gl}_n$, as was first observed by Ragoucy and Sorba. In the general case, they introduced some generalizations of the Yangian which is the so-called shifted Yangians. Inspired by this work, Kang Lu *et al.*[LWW25] constructed a family of truncated shifted twisted Yangians (TSTY) of type AI, denoted by $y^+_{n,ell}(\sigma)$, which was defined very differently (generalizing[BK06]) in terms of generators and relations, Applied the theory of universal equivariant quantizations of conic symplectic singularities to show that W-algebra $y^+_{n,ell}(\sigma)$ and finite W-algebras are isomorphic, quantizing the corresponding Slodowy slices of type BCD.

1.3. **Quantization of Affine Grassmannian.** Let G be a complex semisimple group and consider its thick affine Grassmannian $\mathrm{Gr} = G((t^{-1}))/G[t]$. Attached to each pair of dominant coweights $\lambda \geq \mu$, we have Schubert varieties Gr^λ, $\mathrm{Gr}^\mu \subset \mathrm{Gr}$, with $\mathrm{Gr}^\mu \subset \overline{\mathrm{Gr}^\lambda}$. The neighborhood in $\overline{\mathrm{Gr}^\lambda}$ of a point in Gr^μ is encapsulated in a transversal slice to the latter variety in the former, which we denote by $\mathrm{Gr}^{\overline{\lambda}}_\mu$. This slice is an important object of study in geometric representation theory because under the geometric Satake correspondence it is related to the μ weight space in the irreducible representation of $G^\vee$ of highest weight λ.

The Manin triple $(\mathfrak{g}[t],\, t^{-1}\mathfrak{g}[[t^{-1}]],\, \mathfrak{g}((t^{-1})))$ provides Gr with the structure of a Poisson variety. The slice Gr^λ_μ is an affine Poisson subvariety, and

thus its coordinate ring is naturally a Poisson algebra. [Kam+14] explicitly described quantizations of this Poisson algebra.

It is known that W-algebras are quantizations of Slodowy slices. On the other hand, by the work [MV08], we have an isomorphism between Slodowy slices for sl_m and slices in the affine Grassmannian for GL_n. Thus via these results, we see that quotients of shifted Yangians for gl_n quantize slices in the affine Grassmannian for GL_n. This motivated us to look for a direct construction of quantizations of affine Grassmannian slices (for any semisimple G) using quotients of shifted Yangians. In [WWY20], The authors presented a quantization of an isomorphism of Mirković and Vybornov which relates the intersection of a Slodowy slice and a nilpotent orbit closure in $\mathfrak{gl}_N$ to a slice between spherical Schubert varieties in the affine Grassmannian of PGL_n (with weights encoded by the Jordan types of the nilpotent orbits). A quantization of the former variety is provided by a parabolic W-algebra and of the latter by a truncated shifted Yangian.

2. Preliminary

2.1. Algebraic Group.

Definition 2.1. An *algebraic group* is an algebraic variety G which is also a group such that the maps defining the group structure $\mu : G \times G \to G$ with $\mu(x, y) = xy$ and $i : x \mapsto x^{-1}$ are morphisms of varieties. (We may view the set of points of the variety $G \times G$ as a product set). If the underlying variety is affine, G is a *linear algebraic group*.

Definition 2.2. We call a connected algebraic group of positive dimension *semisimple* if it has no closed connected commutative normal subgroup except e.

2.2. Symplectic Geometry.

[CG10, Chaper 1][TA16, Part 3] Let M be a closed subgroup of an semisimple algebraic group G We first introduce the coadjoint action. The conjugation action of M on G is given by

$$\mathrm{AD} : M \to \mathrm{GL}(G), \qquad m \mapsto \mathrm{AD}_m,$$

where

$$\mathrm{AD}_m(g') = mg'm^{-1}.$$

Differentiating at the identity, we obtain the adjoint action

$$\mathrm{Ad}_m : \mathfrak{g} \to \mathfrak{g}, \qquad g \mapsto \left.\frac{d}{dt}\right|_{t=0} \big(m \exp(tg) m^{-1}\big).$$

Taking the dual map yields

$$\mathrm{Ad}_m^* : \mathfrak{g}^* \to \mathfrak{g}^*.$$

Hence M acts on $\mathfrak{g}^*$ via the coadjoint action:

$$M \to \mathrm{GL}(\mathfrak{g}^*), \qquad m \mapsto \mathrm{Ad}_m^*.$$

Proposition 2.3 (*Kirillov–Kostant–Souriau* symplectic structure)**.** *Any coadjoint orbit $\mathcal{O} \subset \mathfrak{g}^*$ has a natural symplectic structure as below.*

Pick a point $\alpha \in \mathcal{O} \subset \mathfrak{g}^*$. We have a skew-symmetric form on $T_\alpha \mathcal{O} \simeq \mathfrak{g}/\mathfrak{g}^\alpha$, where $\mathfrak{g}^\alpha = \mathrm{Lie}\, G^\alpha$.

We define first a skew-symmetric form

$$\omega_\alpha \colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}, \qquad \omega_\alpha(x, y) \mapsto \alpha([x, y]).$$

And descends to $\mathfrak{g}/\mathfrak{g}^\alpha$ i.e.

$$\omega_\alpha(x, y) = 0 \quad \forall\, y \in \mathfrak{g} \iff x \in \mathfrak{g}^\alpha.$$

ω_α is also non-degenerate, hence ω_α defines a symplectic form on $\mathcal{O}$. And the assignment $\alpha \mapsto \omega_\alpha$ gives a closed 2-form ω on $\mathcal{O}$ which is a symplectic form.

Let A be a commutative, associative unital $\mathbb{C}$-algebra with multiplication $\cdot \colon A \times A \to A$.

Definition 2.4. A commutative algebra $(A, \cdot)$ endowed with an additional $\mathbb{C}$-bilinear anti-symmetric bracket $\{\,,\,\} \colon A \times A \to A$ is called a *Poisson algebra* if the following conditions hold:

(1) A is a Lie algebra with respect to $\{\,,\,\}$;
(2) Leibniz rule:

$$\{a, b \cdot c\} = \{a, b\} \cdot c + b \cdot \{a, c\}, \qquad \forall a, b, c \in A.$$

The Lie bracket $\{\,,\,\}$ will be called a *Poisson bracket* on A. We say that $\{\,,\,\}$ gives a *Poisson structure* on the commutative algebra $(A, \cdot)$.

We will obtain a Poisson structure arising from symplectic structure called the Lie–Poisson bracket. Let (M, ω) be a symplectic variety. The non-degenerate 2-form ω gives a canonical isomorphism $TM \simeq T^*M$. Define a unique $\mathbb{C}$-linear map

$$\mathcal{O}(M) \to \{\text{vector fields on } M\}, \quad f \mapsto \xi_f,$$

by the requirement

$$\omega(\,\cdot\,, \xi_f) = df.$$

Observe that for any vector field η and any function f, by definition of ξ_f we have

$$\omega(\xi_f, \eta) = -\eta f.$$

We define a bracket on $\mathcal{O}(M)$ by any of the following equivalent expressions:

$$\{f, g\} = \omega(\xi_f, \xi_g) = -\xi_g f = \xi_f g.$$

Definition 2.5 (Reduced Schemes)**.** A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subseteq X$, $\mathcal{O}_X(U)$ has no nilpotent elements.

Definition 2.6. An *affine Poisson scheme* (resp. *affine Poisson variety*) is an affine scheme $X = \operatorname{Spec} A$ (resp. $X = \operatorname{Specm} A$) such that A is a Poisson algebra. A *Poisson scheme* (resp. *Poisson variety*) is a scheme (resp. reduced scheme) such that the structure sheaf $\mathcal{O}_X$ is a sheaf of Poisson algebras.

Definition 2.7. A Hamiltonian flow in X from x to x' is a smooth curve $\gamma\colon [0,1] \to X$, defined on an open neighborhood of $[0,1]$ in $\mathbb{C}$, such that $\gamma(0) = x$, $\gamma(1) = x'$, and γ is an integral curve of a Hamiltonian vector field $\xi_f = \{f, \cdot\}$ for some $f \in \mathcal{O}(X)$, defined on an open neighborhood of $\gamma([0,1])$.

2.3. **Algebraic Geometry.**

Definition 2.8. Let X be an algebraic variety over $\mathbb{C}$, and let $x \in X$. We say that X is *smooth at* x if the local ring $\mathcal{O}_{X,x}$ is a regular local ring. Equivalently,

$$\dim_{\mathbb{C}}(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim X,$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. If X is smooth at every point $x \in X$, then X is called a *smooth algebraic variety over* $\mathbb{C}$.

Definition 2.9. Let X be a smooth algebraic variety over $\mathbb{C}$, and let $x \in X$. Write $\mathcal{O}_{X,x}$ for the local ring at x, and $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$ for its maximal ideal. The tangent space of X at x is defined by

$$T_x X := \mathrm{Hom}_{\mathbb{C}}\big(\mathfrak{m}_x/\mathfrak{m}_x^2, \mathbb{C}\big).$$

Example 2.10. If $f : X \to Y$ is a morphism of smooth varieties and $x \in X$, then the differential

$$df_x : T_x X \to T_{f(x)} Y$$

is given on derivations by

$$(df_x(\xi))(g) := \xi(g \circ f), \qquad \xi \in T_x X, \quad g \in \mathcal{O}_{Y,f(x)}.$$

Let G be a connected algebraic group with Lie algebra $\mathfrak{g}$. Consider the adjoint orbit map

$$\mu_f : G \to \mathfrak{g}, \qquad g \mapsto \mathrm{Ad}(g) f.$$

Since $\mathfrak{g}$ is a vector space, we identify

$$T_f \mathfrak{g} \cong \mathfrak{g}.$$

Also,

$$T_1 G \cong \mathfrak{g}.$$

For $x \in \mathfrak{g}$, take the curve $g(t) = \exp(tx)$. Then

$$d\mu_f|_1(x) = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp(tx)) f = \frac{d}{dt}\bigg|_{t=0} e^{t\,\mathrm{ad}\,x}(f) = [x, f].$$

Hence the tangent space to the adjoint orbit $G \cdot f$ at the point f is

$$T_f(G \cdot f) = \mathrm{Im}(d\mu_f|_1) = [\mathfrak{g}, f] = (\mathrm{ad}\, f)(\mathfrak{g}).$$

2.4. **Clifford Algebra.** We will introduce superalgebras especially Clifford algebras.

Definition 2.11. A superspace is a $\mathbb{C}$-vector space E equipped with a $\mathbb{Z}_2$-grading, $E = E^{\bar{0}} \oplus E^{\bar{1}}$. Elements in $E^{\bar{0}}$ are called *even*, elements of $E^{\bar{1}}$ are called *odd*. A superspace which is also an algebra is called *superalgebra*.

Example 2.12. Let E be a $\mathbb{C}$-vector space. The *exterior algebra* $\wedge E$ is the quotient of the tensor algebra $T(E) = \bigoplus_{k \in \mathbb{Z}} T(E)^k$, with $T^k(E) = E \otimes \cdots \otimes E$ the k-fold tensor product, by the two-sided ideal $I(E)$ generated by elements of the form $v \otimes w + w \otimes v$ with $v, w \in E$. The product in

$\wedge E$ is usually denoted by $v \wedge w$. Since $I(E)$ is graded, the exterior algebra $\mathbb{Z}$-grading

$$\wedge(E) = \bigoplus_{k\in\mathbb{Z}} \wedge^k(E).$$

We will regard $\wedge(E)$ as a graded superalgebra, where the $\mathbb{Z}_2$-grading is the mod 2 reduction of the $\mathbb{Z}$-grading.

Definition 2.13. The Clifford algebra $Cl(E,B)$ is the quotient of $T(E)$ by the two-sided ideal $\mathcal{I}(E,B)$ generated by all elements of the form

$$v \otimes w + w \otimes v - B(v,w)1, \quad v, w \in E.$$

Let us view $T(E) = \bigoplus_{k\in\mathbb{Z}} T^k(E)$ as a filtered superalgebra, with the $\mathbb{Z}_2$-grading and filtration inherited from the $\mathbb{Z}$-grading. Since the elements $v \otimes w + w \otimes v - B(v,w)1$ are even, of filtration degree 2, the ideal $\mathcal{I}(E,B)$ is a filtered super subspace of $T(E)$, and hence $Cl(E,B)$ inherits the structure of a filtered superalgebra.

The $\mathbb{Z}_2$-grading and filtration on $Cl(E,B)$ are defined by the condition that the generators $v \in E$ are odd, of filtration degree 1. In the decomposition

$$Cl(E,B) = Cl(E,B)^{\bar{0}} \oplus Cl(E,B)^{\bar{1}}$$

the two summands are spanned by products $v_1 \dots v_k$ with k even, respectively odd. We will always regard $Cl(E,B)$ as a filtered superalgebra. Then the defining relations for the Clifford algebra become

$$[v,w] = vw + wv = B(v,w), \qquad v, w \in E.$$

The map given by the anti-symmetrization:

$$q\colon \wedge(E) \to Cl(E,B), \quad v_1 \wedge \dots \wedge v_k \mapsto \sum_{\sigma\in\mathfrak{S}_k} \operatorname{sgn}(\sigma) v_{\sigma 1} \dots v_{\sigma k},$$

with $\mathfrak{S}_k$ the permutation group of order k, is an isomorphism of superspaces. Its inverse is called the *symbol map*.

Proposition 2.14. *The symbol map $\sigma\colon Cl(E,B) \to \wedge(E)$ induces an isomorphism of graded superalgebras,*

$$\operatorname{gr} Cl(E,B) \xrightarrow{\simeq} \wedge(E).$$

$\operatorname{gr} Cl(E,B)$ inherits a Poisson superalgebra structure, and the symbol map is an isomorphism of graded Poisson superalgebras. The Poisson bracket on $\wedge(E)$ can be described by:

$$\{v,w\} = B(v,w), \quad v,w \in E = \wedge^1(E).$$

2.5. **GIT Quotient.** [New09], [Kir16, §9.2, 9.3]

We claim without proof that any linear algebraic group G over k contains a unique maximal connected normal solvable subgroup, called the *radical* of G.

Definition. Let G be a linear algebraic group. Then G is *reductive* if its radical is isomorphic to a direct product of copies of k^*.

Definition 2.15. Let G be a reductive algebraic group acting algebraically on an affine algebraic variety M. Then the affine quotient of M by G is

$$M//G := \operatorname{Spec}\big(k[M]^G\big).$$

Let χ be a character of G, i.e. a morphism of algebraic groups

$$\chi\colon G \to \mathbf{k}^\times.$$

Definition 2.16 (Twisted GIT quotient)**.** Let G be a reductive algebraic group acting algebraically on an affine algebraic variety M, and let

$$\chi : G \to k^\times$$

be a character of G. For each $n \geq 0$, define

$$k[M]^{G,\chi^n} := \{\, f \in k[M] \mid f(g\cdot m) = \chi(g)^n f(m) \text{ for all } g \in G,\ m \in M \,\}.$$

Set

$$A_\chi(M) := \bigoplus_{n\geq 0} k[M]^{G,\chi^n}.$$

The χ-twisted GIT quotient of M by G is

$$M//_\chi G := \operatorname{Proj}\big(A_\chi(M)\big) = \operatorname{Proj}\left(\bigoplus_{n\geq 0} k[M]^{G,\chi^n}\right).$$

Its degree-zero part is

$$A_\chi(M)_0 = k[M]^G,$$

hence there is a natural projective morphism

$$\pi : M//_\chi G \to M//G, \qquad M//G := \operatorname{Spec}(k[M]^G).$$

3. Slodowy Slice and Its Quantization

In this section we mainly follow the notations in [Slo80], [TA16] and [GG02].

3.1. **Jacobson-Morozov theorem and Slodowy slice.** Let $\mathfrak{sl}_2 = ke + kh + kh$ with the elements

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

which satisfy the following commutation relations:

$$[h, e] = 2e, \ [h, f] = -2f, \ [e, f] = h.$$

Definition 3.1. representation $\rho : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ of $\mathfrak{sl}_2$ on a finite-dimensional vector space V is called *good*, when either $char(k) = 0$ or $char(k) = p > 0$ and $\rho(E)^{p-1} = \rho(F)^{p-1} = 0$.

Definition 3.2. Suppose that A is endowed with an additional $\mathbb{C}$-bilinear bracket $\{\ ,\ \} : A \times A \to A$. Then A is called a *Poisson algebra* if the following conditions hold:

(1) A is a Lie algebra with respect to $\{\ ,\ \}$,
(2) (Leibniz rule) $\{a, b \cdot c\} = \{a, b\} \cdot c + b \cdot \{a, c\}$, for all $a, b, c \in A$.
(3) The Lie bracket $\{\ ,\ \}$ is called a *Poisson bracket* on A.

Let the d-dimensional vector space V_d be spanned by the basis elements $v_1, \ldots, v_d$. If $\mathrm{char}(k) = p > 0$, let $d < p$ otherwise we don't have restriction on d. A good, d-dimensional, irreducible representation of $\mathfrak{sl}_2$ is defined by:

$$\begin{aligned} \rho_d(E)v_1 &= \rho_d(F)v_d = 0 \\ \rho_d(E)v_{i+1} &= \overline{i(d-i)}\, v_i \quad \text{for } 1 \le i \le d-1 \\ \rho_d(F)v_i &= v_{i+1} \quad \text{for } 1 \le i \le d-1 \\ \rho_d(H)v_i &= \overline{(d-2i+1)}\, v_i \end{aligned}$$

(where for $n \in \mathbb{Z}$, $\overline{n}$ is the residue class modulo p).

The next theorem is the key to define Slodowy slice. Let $\mathfrak{g}$ be the Lie algebra of the semisimple algebraic group G (of type Δ)[1] and let $x \in \mathcal{N}(\mathfrak{g})$ be a nilpotent element in $\mathfrak{g}$.

Definition 3.3. A triple (e, h, f) of elements of $\mathfrak{g}$ is called an $\mathfrak{sl}_2$-triple for (the nilpotent element) e, when there is a homomorphism of Lie algebras $\rho : \mathfrak{sl}_2 \to \mathfrak{g}$ with $\rho(E) = e$, $\rho(H) = h$, $\rho(F) = f$.

The *Coxeter number* $\mathrm{Cox}(G)$ or $\mathrm{Cox}(\Delta)$ of G or Δ is $m + 1$, where m is the height of the largest root in the root system belonging to G. We have the following values for $\mathrm{Cox}(G)$:

type of G	A_r	B_r	C_r	D_r	E_6	E_7	E_8	F_4	G_2
$\mathrm{Cox}(G)$	$r+1$	$2r$	$2r$	$2r-2$	12	18	30	12	6

Theorem 3.4 (Jacobson-Morozov)**.** *Let* $\mathrm{char}(k) = 0$ *or* $> 4\,\mathrm{Cox}(G) - 2$. *For every nilpotent element* $e \in \mathfrak{g}$ *there exists an* $\mathfrak{sl}_2$*-triple, and the composition of the associated representation* $\rho : \mathfrak{sl}_2 \to \mathfrak{g}$ *with the adjoint representation* $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ *is good.*

In this paper we mainly focus on the $\mathfrak{sl}_n$ whose corresponding group is SL_n. Every nilpotent matrix in $\mathfrak{sl}_n$ is conjugate to a Jordan block diagonal matrix. Therefore, the nilpotent orbits in $\mathfrak{g}$ are parameterized by $\mathcal{P}(n)$. Thus the nilpotent orbits in $\mathfrak{sl}_n$ corresponds to a set in $\mathcal{P}(n)$ of partitions of n.

Remark 3.5. Let $n \in \mathbb{Z}_{>0}$. We write an element $\boldsymbol{\lambda}$ of partition of n, denote by $\mathcal{P}(n)$, as a decreasing sequence $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_s)$ omitting the zeroes. Thus,

$$\lambda_1 \geq \cdots \geq \lambda_s \geq 1 \quad \text{and} \quad \lambda_1 + \cdots + \lambda_s = n.$$

We shall denote the dual partition of a partition $\boldsymbol{\lambda} \in \mathcal{P}(n)$ by ${}^t\boldsymbol{\lambda}$ whose young tableau is the transpose of $\boldsymbol{\lambda}$.

Example 3.6. Let $\mathfrak{g} = \mathfrak{sl}_n$, and set,

$$e := J_n, \qquad h := \sum_{i=1}^{n} (n + 1 - 2i) e_{i,i}, \qquad f := \sum_{i=1}^{n-1} i(n-i) e_{i+1,i}.$$

[1] we know that $\mathfrak{g}$ is semisimple by [Hum75, § 13.5]

Then (e, h, f) is an $\mathfrak{sl}_2$-triple. From this observation, we readily construct $\mathfrak{sl}_2$-triples for any standard Jordan form $\mathrm{diag}(J_{\lambda_1}, \ldots, J_{\lambda_n})$ with $(\lambda_1, \ldots, \lambda_n) \in \mathcal{P}(n)$.

For the partition $(3, 1)$, we obtain

$$e = E_{12} + E_{23}, \qquad h = \mathrm{diag}(2, 0, -2, 0), \qquad f = 2E_{21} + 2E_{32}.$$

One checks directly that

$$[h, e] = 2e, \qquad [e, f] = h, \qquad [h, f] = -2f.$$

We then state a theorem by [Jac58]:

Theorem 3.7. *Every good representation ρ of $\mathfrak{sl}_2$ is completely reducible and decomposes into a direct sum of irreducible good representations of type ρ_d, $d \in \mathbb{N}^+$, ($d < p$ when $\mathrm{char}(k) = p > 0$). In particular, for every natural number $d \in \mathbb{N}^+$, with $d < p$ if $\mathrm{char}(k) = p > 0$, there exists exactly one irreducible representation of dimension d.*

On the vector space V_d of the good representation ρ_d we can define an action of the multiplicative group k^* by linearly extending the rule

$$(t, v_i) \mapsto t^{d-2i+1} \cdot v_i\,, \quad t \in k^*\,,\ 1 \leq i \leq d\,.$$

If $\rho : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ is any good representation, then we obtain a uniquely defined k^*-action $\lambda : k^* \to \mathrm{GL}(V)$ on V by letting k^* operate on each irreducible component of type V_d in the way above. For the differential $D_e\lambda : \mathrm{Lie}(k^*) \to \mathfrak{gl}(V)$ we then have $D_e(\lambda)(1) = \rho(H)$ (identifying $\mathrm{Lie}(k^*)$ with k).

An element $v \in V$ of a good representation is called an *eigenvector of* λ or *of* H *of integral weight* $n \in \mathbb{Z}$ when

$$\lambda(t)v = t^n v \quad , \quad \text{for all} \quad t \in k^*\,.$$

For $\mathrm{char}(k) = 0$ this condition is equivalent to the infinitesimal condition

$$\rho(H)v = nv\,.$$

In the following we will understand weights of H as always being integral weights in the above sense. For any $n \in \mathbb{Z}$ let $V(n) = \{v \in V \mid \lambda(t)v = t^n v\}$ be the eigenspace of weight n.

Let $V = V_d$ be the vector space of the good representation ρ_d. Then V decomposes into the direct sum of d one-dimensional eigenspaces $V(n)$, $n = d-1, d-3, \ldots, -(d-3), -(d-1)$, and the nilpotent endomorphism $\rho_d(E)$ (resp. $\rho_d(F)$) induces isomorphisms $E : V(n) \xrightarrow{\sim} V(n+2)$ for $n < d-1$ (resp. $F : V(n) \xrightarrow{\sim} V(n-2)$ for $n > -(d-1)$) and annihilates $V(d-1)$ (resp. $V(-(d-1))$). Below we will only focus on cases when $\mathrm{char}(k) = 0$.

Definition 3.8. Let V be a smooth variety, let $\mathcal{O} \subset V$ be a smooth subvariety, and let $y \in \mathcal{O}$. A subvariety $S \subset V$ passing through y is called a *transverse slice to $\mathcal{O}$ at y* if

$$T_y V = T_y \mathcal{O} \oplus T_y S.$$

Let (e, h, f) be an $\mathfrak{sl}_2$-triple in $\mathfrak{g}$. Consider the orbit map

$$\mu_f : G \to \mathfrak{g}, \qquad g \mapsto \mathrm{Ad}(g) f.$$

Its differential at the identity element $1 \in G$ is given by

$$d\mu_f|_1 : \mathfrak{g} \to \mathfrak{g}, \qquad x \mapsto [x, f] = (\mathrm{ad}\, f)(x).$$

Hence the tangent space to the adjoint orbit $G \cdot f$ at the point f is

$$T_f(G \cdot f) = (\mathrm{ad}\, f)(\mathfrak{g}).$$

By Theorem 3.7, the adjoint representation of the $\mathfrak{sl}_2$-triple decomposes as

$$\mathfrak{g} = \bigoplus_{j=1}^{s} E_j,$$

where each E_j is an irreducible $\mathfrak{sl}_2$-module of dimension $n_j + 1$. Writing

$$E_j = \bigoplus_{m=-n_j}^{n_j} E_j(m)$$

for its weight space decomposition, each $E_j(m)$ is one-dimensional, and for every

$$m \leq n_j - 2$$

the map $\mathrm{ad}\, f$ restricts to an isomorphism

$$\mathrm{ad}\, f : E_j(m+2) \xrightarrow{\sim} E_j(m).$$

Therefore,

$$(\operatorname{ad} f)(E_j) = \bigoplus_{m=-n_j}^{n_j-2} E_j(m),$$

so the highest weight space $E_j(n_j)$ is a complement to $(\operatorname{ad} f)(E_j)$ in E_j. Since

$$E_j(n_j) = \ker(\operatorname{ad} e) \cap E_j,$$

we obtain

$$\ker(\operatorname{ad} e) = \bigoplus_{j=1}^{s} E_j(n_j).$$

Consequently,

$$\mathfrak{g} = (\operatorname{ad} f)(\mathfrak{g}) \oplus \ker(\operatorname{ad} e).$$

Definition 3.9. The Slodowy slice through f is the affine subspace

$$\mathcal{S}_f := f + \ker(\operatorname{ad} e) = f + \mathfrak{g}^e,$$

Proposition 3.10. *The Slodowy slice $\mathcal{S}_f$ is transverse to the adjoint orbit $G \cdot f$ at the point f. More precisely,*

$$T_f\mathfrak{g} = T_f(G \cdot f) \oplus T_f(\mathcal{S}_f).$$

Proof. Since $\mathcal{S}_f = f + \mathfrak{g}^e$ is an affine subspace of $\mathfrak{g}$, its tangent space at f is

$$T_f(\mathcal{S}_f) = \mathfrak{g}^e = \ker(\operatorname{ad} e).$$

On the other hand, from the differential of the orbit map we have

$$T_f(G \cdot f) = (\operatorname{ad} f)(\mathfrak{g}).$$

Using the decomposition established above,

$$\mathfrak{g} = (\operatorname{ad} f)(\mathfrak{g}) \oplus \ker(\operatorname{ad} e),$$

we conclude that

$$T_f\mathfrak{g} = \mathfrak{g} = T_f(G \cdot f) \oplus T_f(\mathcal{S}_f).$$

Thus $\mathcal{S}_f$ is transverse to the orbit $G \cdot f$ at f. □

In particular, $\mathcal{S}_f$ defines a transverse slice to the adjoint orbit of f at the point f.

Example 3.11. Let $Y = (y_{ij}) \in \mathfrak{sl}_4$. Solving

$$[Y, e] = 0, \qquad e = E_{12} + E_{23},$$

gives

$$Y = \begin{pmatrix} a & b & c & u \\ 0 & a & b & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix}.$$

Therefore

$$\mathfrak{g}^e = \left\{ \begin{pmatrix} a & b & c & u \\ 0 & a & b & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix} \middle| \, a, b, c, u, v \in \mathbb{C} \right\},$$

which indeed has dimension 5.

Hence

$$\mathcal{S}_f = f + \mathfrak{g}^e = \left\{ \begin{pmatrix} a & b & c & u \\ 2 & a & b & 0 \\ 0 & 2 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix} \middle| \, a, b, c, u, v \in \mathbb{C} \right\}.$$

Thus, as an affine variety,

$$\mathcal{S}_f \cong \mathbb{A}^5$$

with coordinate ring

$$\mathbb{C}[\mathcal{S}_f] \cong \mathbb{C}[a, b, c, u, v].$$

3.2. **Nilpotency.**

Definition 3.12. The *nilpotent cone* of $\mathfrak{g}$, denoted by $\mathcal{N}$, is the set of all nilpotent elements of $\mathfrak{g}$.

For $e \in \mathfrak{g}$, we denote by $G.e$ its adjoint G-orbit. The nilpotent cone is a finite union of nilpotent G-orbits. An element $x \in \mathfrak{g}$ is *regular* if its centralizer $\mathfrak{g}^x$ has the minimal dimension, that is, the rank r of $\mathfrak{g}$. The maximal dimensional orbit $\mathcal{O}_{\text{reg}}$ is the set of all regular nilpotent elements of $\mathfrak{g}$.

Note that If $\mathfrak{g}$ is a simple Lie algebra of matrices, $\mathcal{N}$ coincides with the set of nilpotent matrices by preservation of Jordan decomposation [Hum72, section 6.3].

Example 3.13. In $\mathfrak{sl}_n$, regular nilpotent elements are just conjugacy class of the n-size Jordan nilpotent Jordan blocks.

$$e = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Matrices that commute with e in $\mathfrak{gl}_n$ are polynomials of e:

$$a_0 I + a_1 e + \cdots + a_{n-1} e^{n-1},$$

thus dimension is n.

Quotient the trace 0 condition in $\mathfrak{sl}_n$, the generator I will be canceled. Thus

$$\dim \mathfrak{g}^e = n - 1 = \operatorname{rank} \mathfrak{sl}_n.$$

Next, there is a unique maximal dimensional orbit in $\mathcal{N} \setminus \mathcal{O}_{\text{reg}}$ which is called the *subregular nilpotent orbit* of $\mathfrak{g}$, and denoted by $\mathcal{O}_{\text{subreg}}$. Its codimension in $\mathfrak{g}$ is the rank of $\mathfrak{g}$ plus two.

Example 3.14. Consider The nilpotent Jordan block with partition $(n-1, 1)$, that is, with one $(n-1)$-size Jordan block and one 1-size Jordan block, denote by X.

$$\mathfrak{g}^X = \{Y \in \mathfrak{sl}_n : [X, Y] = 0\}.$$

Let

$$Y = \begin{pmatrix} A & b \\ c & d \end{pmatrix},$$

$A \in M_{n-1}((n-1) \times (n-1)$ matrix), $b \in M_{n-1,1}$(column vector), $c \in M_{1,n-1}$(row vector), $d \in \mathbb{C}$.

$$X = \begin{pmatrix} J & 0 \\ 0 & 0 \end{pmatrix}, \quad J := J_{n-1},$$

where J_{n-1} is $(n-1)$-size nilpotent Jordan block.

Thus

$$XY = \begin{pmatrix} JA & Jb \\ 0 & 0 \end{pmatrix}, \quad YX = \begin{pmatrix} AJ & 0 \\ cJ & 0 \end{pmatrix}.$$

Then we have

$$JA = AJ, \quad Jb = 0, \quad cJ = 0.$$

Thus

$$c = (0, \dots, 0, c_{n-1}), \quad b = (b_1, 0, \dots, 0)^T, \quad A = a_0 I + a_1 J + \cdots + a_{n-2} J^{n-2}.$$

Since $\operatorname{tr} Y = 0$, d is determined by A. Thus the dimension is uniquely determined by above result, which is $1+1+n-1 = n-1+2 = \operatorname{rank} \mathfrak{sl}_{\mathfrak{n}} + 2$.

Particularly we focus on $\mathfrak{sl}_4$ for its subregular cases, that is $(3,1)$ partition of 4, most of this section. And we will not distinguish $\mathfrak{g}$ and $\mathfrak{sl}_4$ any more.

Example 3.15. We shall denote by $\mathcal{O}_{\boldsymbol{\lambda}}$ the corresponding nilpotent orbit of $\mathfrak{sl}_n$. Then $\mathcal{O}_{\boldsymbol{\lambda}}$ is represented by the standard Jordan form $\operatorname{diag}(J_{\lambda_1}, \dots, J_{\lambda_s})$, where J_k is the k-size Jordan block. If we write ${}^t\boldsymbol{\lambda} = (d_1, \dots, d_t)$, one can prove that for some $e \in \mathcal{O}_{\boldsymbol{\lambda}}$

$$\dim \mathfrak{g}^e = \sum_i d_i^2 - 1.$$

So

$$\dim \mathcal{O}_{\boldsymbol{\lambda}} = n^2 - \sum_{i=1}^{t} d_i^2.$$

For $\boldsymbol{\lambda} = (3,1)$ in $\mathfrak{sl}_4$, since ${}^t(3,1) = (2,1,1)$, we get

$$\dim \mathcal{O}_{(3,1)} = 4^2 - (2^2 + 1^2 + 1^2) = 16 - 6 = 10.$$

Hence

$$\dim \mathfrak{g}^e = \dim \mathfrak{sl}_4 - \dim \mathcal{O}_{(3,1)} = 15 - 10 = 5.$$

Recall that by Jacobson-Morozov theorem 3.4 we have $\mathfrak{sl}_2$-triple e, f, h.

Definition 3.16. Since the eigenvalues of $\operatorname{ad} h$ are integers, we get a $\frac{1}{2}\mathbb{Z}$-grading on $\mathfrak{g}$ defined by h, called the *Dynkin grading associated with* h:

$$\mathfrak{g} = \bigoplus_{j \in \frac{1}{2}\mathbb{Z}} \mathfrak{g}_j, \qquad \mathfrak{g}_j := \{x \in \mathfrak{g} \mid [h, x] = 2jx\}.$$

We call e an *even nilpotent element* if $\mathfrak{g}_i = \{0\}$ for all half-integers i. The nilpotent orbit of an even nilpotent element is called an *even nilpotent orbit*.

Example 3.17. Since

$$h = \mathrm{diag}(2, 0, -2, 0),$$

Let

$$w_1 = 2, \qquad w_2 = 0, \qquad w_3 = -2, \qquad w_4 = 0,$$

then

$$[h, E_{ij}] = (w_i - w_j)E_{ij}.$$

Hence

$$\mathfrak{g}_2 = \mathbb{C}E_{13},$$

$$\mathfrak{g}_1 = \mathrm{span}\{E_{12}, E_{14}, E_{23}, E_{43}\},$$

$$\mathfrak{g}_0 = \left\{ \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & u \\ 0 & 0 & c & 0 \\ 0 & v & 0 & d \end{pmatrix} \middle| \; a + b + c + d = 0 \right\},$$

$$\mathfrak{g}_{-1} = \mathrm{span}\{E_{21}, E_{41}, E_{32}, E_{34}\}, \qquad \mathfrak{g}_{-2} = \mathbb{C}E_{31}.$$

In particular, this nilpotent element is *even*

Below we will often identify $\mathfrak{g}$ with its dual $\mathfrak{g}^*$ through the non-degenerate invariant bilinear form $(\ |\ )$. Let (e, h, f) be an $\mathfrak{sl}_2$-triple of $\mathfrak{g}$ and let $\chi = (f|\cdot)$ be the linear form associated with f. Particularly in the semisimple type A case the non-degenerate bilinear could be Killing form.

Proposition 3.18. *The restriction of the antisymmetric bilinear form,*

$$\omega_\chi : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}, \qquad (x, y) \mapsto (f|[x, y]),$$

to $\mathfrak{g}_{\frac{1}{2}} \times \mathfrak{g}_{\frac{1}{2}}$ *is non-degenerate.*

Proof. Its easy to see a bijection $\mathrm{ad}\, f : \mathfrak{g}_{\frac{1}{2}} \to \mathfrak{g}_{-\frac{1}{2}}$ whose inverse is $\mathrm{ad}\, e$. Thus

$$\omega_\chi(x, y) = (f \mid [x, y]) = ([f, x] \mid y),$$

since the bilinear form is invariant under Lie bracket. Using the fact that $(\ |\ )$ is non-degenerate on $\mathfrak{g}_{\frac{1}{2}} \times \mathfrak{g}_{-\frac{1}{2}}$ If $\omega_\chi(x, y) = 0$ for all y, then $[f, x] = 0$ because above $\mathrm{ad}\, f$ is injective, which implies $x = 0$. Hence, this directly ensures that ω_χ is nondegenerate on $\mathfrak{g}_{\frac{1}{2}}$.

□

ω_χ is called the *Kirillov form associated with* f. Let $\mathcal{L}$ be a Lagrangian subspace of $\mathfrak{g}_{\frac{1}{2}}$, that is, $\omega_\chi(\mathcal{L},\mathcal{L}) = 0$ and $\dim \mathcal{L} = \frac{1}{2}\dim \mathfrak{g}_{\frac{1}{2}}$, and set

$$\mathfrak{m} = \mathfrak{m}_{\chi,\mathcal{L}} := \mathcal{L} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j. \tag{1}$$

Then $\mathfrak{m}$ is an ad-nilpotent (since $[\mathfrak{g}_i, \mathfrak{g}_j] \subset \mathfrak{g}_{i+j}$), $\operatorname{ad} h$-graded subalgebra of $\mathfrak{g}$. And $\chi([\mathfrak{m},\mathfrak{m}]) = (f|[\mathfrak{m},\mathfrak{m}]) = 0$ since $(\mathfrak{g}_i \mid \mathfrak{g}_j) = 0$ unless $i+j=0$ i.e. χ is a character on $\mathfrak{m}$.

Example 3.19. In our case

$$\chi = (f \mid \cdot), \qquad f = 2E_{21} + 2E_{32}.$$

Since by the well known result $k(A,B) := \operatorname{tr}(\operatorname{ad} A \cdot \operatorname{ad} B) = 2n \operatorname{tr}(AB)$. We just need to compute the trace pairing,

$$\chi(E_{12}) = 2, \qquad \chi(E_{23}) = 2, \qquad \chi(E_{14}) = 0, \qquad \chi(E_{43}) = 0, \qquad \chi(E_{13}) = 0.$$

And since

$$\mathfrak{g}_{1/2} = 0,$$

we have $\mathcal{L} = 0$, hence

$$\mathfrak{m} = \mathfrak{g}_1 \oplus \mathfrak{g}_2.$$

Therefore

$$\mathfrak{m} = \operatorname{span}\{E_{12}, E_{14}, E_{23}, E_{43}, E_{13}\}.$$

3.3. **Posisson structure on $\mathcal{S}_f$.** First, let's consider the dual realization of the Slodowy slice. Let

$$\chi := (f \mid \cdot) \in \mathfrak{g}^*,$$

we define

$$\mathcal{S}_f := \chi + (\mathfrak{g}^f)^* \subset \mathfrak{g}^*.$$

We now show that, via the invariant bilinear form, this affine subspace is naturally identified with the original definition

$$\mathcal{S}_f = f + \mathfrak{g}^e.$$

Proposition 3.20. *Let*

$$\kappa : \mathfrak{g} \xrightarrow{\sim} \mathfrak{g}^*, \qquad x \mapsto (x \mid \cdot)$$

be the identification induced by the invariant bilinear form. Then

$$\kappa\big(f+\mathfrak{g}^e\big)=\chi+(\mathfrak{g}^f)^*.$$

Equivalently,

$$\mathcal{S}_f\cong f+\mathfrak{g}^e.$$

Proof. Since $\kappa(f)=\chi$, it suffices to show that the restriction of κ induces an isomorphism

$$\mathfrak{g}^e\xrightarrow{\sim}(\mathfrak{g}^f)^*.$$

Consider the linear map

$$\varphi:\mathfrak{g}^e\to(\mathfrak{g}^f)^*,\qquad x\mapsto(x\mid\cdot)\big|_{\mathfrak{g}^f}.$$

We first prove that φ is injective. Suppose that $x\in\mathfrak{g}^e$ satisfies

$$(x\mid y)=0\qquad\text{for all } y\in\mathfrak{g}^f.$$

By the $\mathfrak{sl}_2$-module decomposition of $\mathfrak{g}$, similarly as in 3.10 one has

$$\mathfrak{g}=[\mathfrak{g},f]\oplus\mathfrak{g}^e.$$

On the other hand, by invariance of $(\cdot\mid\cdot)$, for any $z\in\mathfrak{g}$ and any $y\in\mathfrak{g}^f$,

$$([z,f]\mid y)=(z\mid[f,y])=0,$$

since $[f,y]=0$. Hence

$$[\mathfrak{g},f]\subseteq(\mathfrak{g}^f)^\perp.$$

Now

$$\dim[\mathfrak{g},f]=\dim\mathfrak{g}-\dim\mathfrak{g}^f=\dim(\mathfrak{g}^f)^\perp,$$

so in fact

$$[\mathfrak{g},f]=(\mathfrak{g}^f)^\perp.$$

Therefore $x\in\mathfrak{g}^e\cap(\mathfrak{g}^f)^\perp=\mathfrak{g}^e\cap[\mathfrak{g},f]$. Since

$$\mathfrak{g}=[\mathfrak{g},f]\oplus\mathfrak{g}^e,$$

this intersection is zero, and hence $x=0$. Thus φ is injective.

Next, we compare dimensions. Since we proved in 3.10 $\ker(\operatorname{ad}e)=\bigoplus_{j=1}^s E_j(n_j)$ and similarly $\ker(\operatorname{ad}f)=\bigoplus_{j=1}^s E_j(-n_j)$, we have

$$\dim\mathfrak{g}^e=\dim\mathfrak{g}^f.$$

Hence

$$\dim \mathfrak{g}^e = \dim(\mathfrak{g}^f)^*.$$

Since φ is an injective linear map between finite-dimensional vector spaces of the same dimension, it is an isomorphism.

Finally, we obtain

$$\kappa\big(f + \mathfrak{g}^e\big) = \chi + (\mathfrak{g}^f)^*.$$

□

To give a generalization of $\mathfrak{m}$. Set $\mathfrak{L}$ to be the isotropic subspace of $\mathfrak{g}_{\frac{1}{2}}$, that is, $\omega_\chi(\mathfrak{L}, \mathfrak{L}) = 0$, and set where

$$\mathfrak{m}_{\chi,\mathcal{L}} := \mathfrak{L} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j, \qquad \mathfrak{n}_{\chi,\mathcal{L}} := \mathcal{L}^{\perp_{\omega_\chi}} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j,$$

and

$$\mathcal{L}^{\perp_{\omega_\chi}} = \left\{ x \in \mathfrak{g}_{\frac{1}{2}} a \mid \omega_\chi(x, \mathcal{L}) = 0 \right\}.$$

Note that $\mathfrak{g}^e \subset \mathfrak{m}^\perp$, since $\mathfrak{g}^e$ consists of all highest weight vectors.

Let $\mathbf{N}$ be the unipotent subgroup of G and hence it is connected [Hum75, Exercise 15.6] with Lie algebra $\mathfrak{n}$. When $\mathfrak{m} = \mathfrak{n}$, then $\mathfrak{L}$ is Lagrangian, since we have $\dim L + \dim L^{\perp\omega} = \dim V$ for all symplectic space (V, ω). We also write $\mathbf{M}$ the closed connected subgroup [Hum75, §13.1] of G with Lie algebra $\mathfrak{m}$. It is by

Consider the adjoint map

$$\mathbf{N} \times (f + \mathfrak{m}^\perp) \longrightarrow \mathfrak{g}, \quad (g, x) \mapsto g \cdot x.$$

Here, $\perp$ denotes orthogonality with respect to the bilinear form $(\ \mid\ )$.

Proposition 3.21. *Its image is contained in $f + \mathfrak{m}^\perp$.*

Proof. Indeed, for any $x \in \mathfrak{n}$ and any $y \in \mathfrak{m}^\perp$, $\exp(\operatorname{ad} x)(f + y) \in f + \mathfrak{m}^\perp$ since $[\mathfrak{n}, \mathfrak{m}] \subset \mathfrak{m}$, $\chi([\mathfrak{n}, \mathfrak{m}]) = 0$. Thus for any $m \in \mathfrak{m}$

$$\begin{aligned}\big(\exp(\operatorname{ad} x)(f + y) - f \big| m\big) &= -(f \big| [f, m]) + \big(\exp(\operatorname{ad} x)(f + y) \big| m\big) \\ &= 0 + \big(\exp(\operatorname{ad} x)(f) \big| m\big) + \big(\exp(\operatorname{ad} y) \big| m\big),\end{aligned}$$

and since $\mathfrak{n}$ being ad-nilpotent

$$\begin{aligned}&= \big(f \big| \exp(\operatorname{ad}(-x))(m)\big) + 0 \\ &= 0.\end{aligned}$$

And this is enough to conclude because $\mathbf{N}$ is generated by the elements $\exp(\operatorname{ad} x)$ for x running through $\mathfrak{n}$. As a result, by restriction, we get a map

$$\alpha\colon \mathbf{N} \times \mathcal{S}_f \longrightarrow f + \mathfrak{m}^{\perp}.$$

□

Thanks to [GG02, §2.3] we have

Theorem 3.22. *The map α is an isomorphism of affine varieties.*

Thus, the induced pullback

$$\alpha^*\colon \mathbb{C}[f + \mathfrak{m}^{\perp}] \longrightarrow \mathbb{C}[\mathbf{N} \times \mathcal{S}_f]$$

is an isomorphism. Consider the α-equivariant action of $\mathbf{N}$ on $\mathbf{N} \times \mathcal{S}_f$ and on $f + \mathfrak{m}^{\perp}$, where the action on $\mathbf{N} \times \mathcal{S}_f$ is given by left translation on the first factor, namely,

$$n \cdot (g, s) := (ng, s), \qquad n, g \in \mathbf{N},\ s \in \mathcal{S}_f,$$

and the action on $f + \mathfrak{m}^{\perp}$ is the adjoint action, that is,

$$n \cdot (g \cdot s) = (ng) \cdot s \qquad \text{for all } n, g \in \mathbf{N},\ s \in \mathcal{S}_f.$$

Now since α is $\mathbf{N}$-equivariant. We have

$$\mathbb{C}[f + \mathfrak{m}^{\perp}]^{\mathbf{N}} \cong \mathbb{C}[\mathbf{N} \times \mathcal{S}_f]^{\mathbf{N}} \cong \mathbb{C}[\mathbf{N}]^{\mathbf{N}} \otimes \mathbb{C}[\mathcal{S}_f].$$

And we have $\mathbb{C}[\mathbf{N}]^{\mathbf{N}} \cong \mathbb{C}$

As a consequence we get:

$$\mathbb{C}[\mathcal{S}_f] \cong \mathbb{C}[f + \mathfrak{m}^{\perp}]^{\mathbf{N}}.$$

Later, we will identify $\mathfrak{m}^{\perp}$ to $\mathfrak{m}^{\circ} := \{\xi \in \mathfrak{g}^* \mid \xi(m) = 0\}$ by $(\ |\ )$.

Let us describe explicitly the Poisson bracket on $\mathbb{C}[\mathfrak{g}^*]$ [CG10, Proposition 1.3.18]. Let $(x_1, \ldots, x_n)$ be a basis of $\mathfrak{g}$, with $[x_i, x_j] = \sum_k c_{i,j}^k x_k$. Through the canonical isomorphism $(\mathfrak{g}^*)^* \cong \mathfrak{g}$, any element of $\mathfrak{g}$ is regarded as a linear function on $\mathfrak{g}^*$, and thus as an element of $\mathbb{C}[\mathfrak{g}^*]$. We have for $f, g \in \mathbb{C}[\mathfrak{g}^*]$,

$$\{f, g\} = \sum_{i,j,k} c_{i,j}^k x_k \frac{\partial f}{\partial x_i^*} \frac{\partial g}{\partial x_j^*}.$$

In a more concise way, we have:

$$\{f,g\}\colon \mathfrak{g}^* \to \mathbb{C}, \quad \xi \mapsto \langle \xi, [d_\xi f, d_\xi g]\rangle \tag{2}$$

where $d_\xi f, d_\xi g \in (\mathfrak{g}^*)^* \cong \mathfrak{g}$ denote the differentials of f and g at $\xi \in \mathfrak{g}^*$. In particular, if $x, y \in \mathfrak{g} \cong (\mathfrak{g}^*)^* \subset \mathbb{C}[\mathfrak{g}^*]$, then

$$\{x, y\} = [x, y].$$

Moreover [CG10, Proposition 1.3.21], for any regular functions $f, g \in \mathbb{C}[\mathfrak{g}^*]$, and any coadjoint orbit $\mathbb{O} \subset \mathfrak{g}^*$ we have

$$\{f, g\}|_{\mathbb{O}} = \{f|_{\mathbb{O}}, g|_{\mathbb{O}}\}_{\text{symplectic}}.$$

The bracket on the right hand side comes from the symplectic structure on $\mathbb{O}$, while the bracket on the left hand side comes from the Poisson bracket on $\mathbb{C}[\mathfrak{g}^*]$ restricted to $\mathbb{O}$.

Definition 3.23. If X is smooth, then one may view X as a complex-analytic manifold equipped with a holomorphic Poisson structure. For each point $x \in X$, one defines the *symplectic leaf* $\mathcal{S}_x$ through x to be the set of points that can be reached from x by flowing along Hamiltonian vector fields.

Example 3.24. The space $\mathfrak{g}^*$ is a (smooth) Poisson variety and the symplectic leaves of $\mathfrak{g}^*$ are the coadjoint orbits of $\mathfrak{g}^*$ [Vai94, Proposition 3.1].

Thanks to [Wei83], we know that the natural Kirillov-Kostant-Souriau Poisson structure on $\mathfrak{g}^*$ can be induced to $\mathcal{S}_f$, that is $\{\ ,\ \}_{\mathcal{S}_f}$ on $\mathbb{C}[\mathcal{S}_f]$ is given by,

$$\{f, g\}_{\mathcal{S}_f}(\xi) = \{f|_{\mathbb{O}}, g|_{\mathbb{O}}\}_{\text{symplectic}}(\xi), \tag{3}$$

for any $f, g \in \mathbb{C}[\mathcal{S}_f]$ and $\xi \in \mathcal{S}_f$, if $\mathbb{O}$ denotes the coadjoint orbit through ξ in $\mathfrak{g}^*$.

But for quantization convenience we introduce another method to obtain Poisson structures by Hamiltonian reduction.

Definition 3.25. The action of a Lie algebra $\mathfrak{a}$ on a manifold X is said to be *Hamiltonian* if there exists a Lie algebra homomorphism

$$\tilde{\mu}\colon \mathfrak{a} \longrightarrow \mathcal{O}_X(X), \qquad a \longmapsto \tilde{\mu}_a,$$

such that the following diagram commutes:

$$\begin{tikzcd} \mathfrak{a} \arrow[r] \arrow[dr, dashed, "\tilde{\mu}"'] & \mathfrak{X}(X) \\ & \mathcal{O}_X(X) \arrow[u] \end{tikzcd}$$

where $\mathfrak{X}(X)$ denotes the Lie algebra of (symplectic) vector fields on X, the vertical arrow is the natural inclusion $\mathcal{O}_X(X) \hookrightarrow \mathfrak{X}(X)$ given by $f \mapsto \xi_f = \{f, \cdot\}$ which is a derivation on $\mathcal{O}_X(X)$, and the horizontal map $\mathfrak{a} \to \mathfrak{X}(X)$ is induced by the $\mathfrak{a}$-action on X, explicitly:

$$a \longmapsto (x \longmapsto \text{the infinitesimal action of } a \curvearrowright x \in T_x X)\,.$$

We call the map $\tilde{\mu}$ the *co-moment map* of the action, or the *Hamiltonian* of the action. Its dual map

$$\mu\colon X \longrightarrow \mathfrak{a}^*, \qquad x \longmapsto \mu(x),$$

defined by $\langle \mu(x), a\rangle = \tilde{\mu}_a(x)$ for all $a \in \mathfrak{a}$, is called the *moment map* of the action.

Remark 3.26. If the group $\mathbf{A}$ is connected, then μ is $\mathbf{A}$-equivariant with respect to the coadjoint action on $\mathfrak{a}^*$.

Theorem 3.27 (Marsden-Weinstein)**.** *Assume that $\mathbf{A}$ is connected and that the action of $\mathbf{A}$ in X is Hamiltonian. Let $\gamma \in \mathfrak{a}^*$. Assume that γ is a regular value*[2] *of μ, that $\mu^{-1}(\gamma)$ is $\mathbf{A}$-stable and that $\mu^{-1}(\gamma)/\mathbf{A}$ is a variety. Let $\iota\colon \mu^{-1}(\gamma) \hookrightarrow X$ and $\pi\colon \mu^{-1}(\gamma) \twoheadrightarrow \mu^{-1}(\gamma)/\mathbf{A}$ be the natural maps: ι is the inclusion and π is the quotient map. Then the triple*

$$(X, \mu^{-1}(\gamma), \mu^{-1}(\gamma)/\mathbf{A})$$

is Poisson-reducible, i.e., there exists a Poisson structure $\{,\}'$ on $\mu^{-1}(\gamma)/\mathbf{A}$ such that for all open subset $U \subset X$ and for all $f, g \in \mathcal{O}_X(\pi(U \cap \mu^{-1}(\gamma)))$, one has

$$\{f, g\}' \circ \pi(u) = \{\tilde{f}, \tilde{g}\} \circ \iota(u)$$

[2]If $f\colon X \to Y$ is a smooth map between varieties, we say that a point y is a *regular value* of f if for all $x \in f^{-1}(y)$, the map $d_x f\colon T_x(X) \to T_y(Y)$ is surjective. If so, then $f^{-1}(y)$ is a subvariety of X and the codimension of this variety in X is equal to the dimension of Y.

at any point $u \in U \cap \mu^{-1}(\gamma)$*, where* $\tilde{f}, \tilde{g} \in \mathcal{O}_X(U)$ *are arbitrary extensions of* $f \circ \pi|_{U\cap\mu^{-1}(\gamma)}, g \circ \pi|_{U\cap\mu^{-1}(\gamma)}$ *to* U*.*

Remark 3.28. The Poisson structure obtained in the theorem is considered obtained by Hamiltonian reduction.

Assume that $\mathfrak{m} = \mathfrak{n}$ and let in this case $\mathbf{M}$ be the unipotent subgroup of $\mathbf{G}$ with Lie algebra $\mathfrak{m}$.

We intend to apply the theorem to the connected Lie group $\mathbf{M}$ acting on the variety $\mathfrak{g}^*$ with The algebra $\mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*)$ is equipped with the Lie–Poisson bracket.

Proposition 3.29. *The action is Hamiltonian and the moment map*

$$\mu\colon \mathfrak{g}^* \to \mathfrak{m}^*$$

is a restriction of functions from $\mathfrak{g}$ *to* $\mathfrak{m}$*.*

Proof. For a regular function $f : \mathfrak{g}^* \to \mathbb{C}$, its differential at $\xi \in \mathfrak{g}^*$ is the linear form

$$df_\xi : T_\xi(\mathfrak{g}^*) \to \mathbb{C}, \qquad df_\xi(v) = \left.\frac{d}{dt}\right|_{t=0} f(\xi + tv).$$

Since $\mathfrak{g}^*$ is a vector space, we identify

$$T_\xi(\mathfrak{g}^*) \cong \mathfrak{g}^*,$$

and hence

$$df_\xi \in (\mathfrak{g}^*)^* \cong \mathfrak{g}.$$

In particular, for the linear function $f_x(\xi) := \xi(x)$ with $x \in \mathfrak{g}$, one has

$$df_{x,\xi} = x.$$

Consider the coadjoint action of M on $\mathfrak{g}^*$. Define

$$\tilde{\mu} : \mathfrak{m} \to \mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*), \qquad \tilde{\mu}_x(\xi) := \xi(x), \quad x \in \mathfrak{m},\ \xi \in \mathfrak{g}^*,$$

which means $d\tilde{\mu}_{x,\xi} = x$. Then

$$\tilde{\mu} : \mathfrak{m} \to \mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*), \qquad x \mapsto \tilde{\mu}_x$$

is the co-moment map. Indeed, by 2, for $x, y \in \mathfrak{m}$,

$$\{\tilde{\mu}_x, \tilde{\mu}_y\}(\xi) = \xi([x, y]) = \tilde{\mu}_{[x,y]}(\xi),$$

that is, $\tilde{\mu}$ is a Lie algebra homomorphism. By definition, the corresponding moment map

$$\mu : \mathfrak{g}^* \to \mathfrak{m}^*$$

is the dual map characterized by

$$\mu(\xi)(x) = \tilde{\mu}_x(\xi) = \xi(x), \qquad x \in \mathfrak{m}.$$

Hence

$$\mu(\xi) = \xi|_{\mathfrak{m}},$$

We now show that the coadjoint action of M on $\mathfrak{g}^*$ is Hamiltonian.

It remains to check that $\tilde{\mu}_x$ generates the infinitesimal coadjoint action of x. For $y \in \mathfrak{g} \cong (\mathfrak{g}^*)^*$, and we don't distinguish y and f_y

$$\begin{aligned}\left(\frac{d}{dt}\bigg|_{t=0} \exp(t\,\mathrm{ad}_x^*)\xi\right)(y) &= \frac{d}{dt}\bigg|_{t=0} \big((\exp(t\,\mathrm{ad}_x^*)\xi)(y)\big) \\ &= (\mathrm{ad}_x^*\,\xi)(y) \\ &= -\xi([x,y]).\end{aligned}$$

On the other hand,

$$\{\tilde{\mu}_x, f_y\}(\xi) = \xi([x,y]).$$

Hence the Hamiltonian vector field of $\tilde{\mu}_x$ coincides with the infinitesimal vector field induced by x. Therefore the coadjoint action of M on $\mathfrak{g}^*$ is Hamiltonian. □

Since $\chi|_{\mathfrak{m}}$ is a character on $\mathfrak{x}$, we have

$$\big(\mathrm{ad}^*\, m(\chi)\big)y = -\chi([x,y]) = 0 \quad x, y \in \mathfrak{m}.$$

Thus

$$\mathrm{ad}^*\, x(\chi|_{\mathfrak{m}}) = 0.$$

That is

$$\mathrm{Ad}^*\, m(\chi|_{\mathfrak{m}}) = \chi|_{\mathfrak{m}} \quad m \in M$$

As a consequence, the set

$$\mu^{-1}(\chi|_{\mathfrak{m}}) = \{\xi \in \mathfrak{g}^* \mid \mu(\xi) = \chi|_{\mathfrak{m}}\}$$

is $\mathbf{M}$-stable. And

$$\mu(\xi) = \chi|_{\mathfrak{m}} \iff (\xi - \chi)|_{\mathfrak{m}} = 0 \iff \xi - \chi \in \mathfrak{m}^\circ \cong \mathfrak{m}^\perp.$$

Therefore,

$$\mu^{-1}(\chi|_{\mathfrak{m}}) = \chi + \mathfrak{m}^{\perp}.$$

Moreover, we have the following lemma:

Lemma 3.30. *[TA16, Lemma 3.23] $\chi|_{\mathfrak{m}}$ is a regular value for the restriction of μ to each symplectic leaf of $\mathfrak{g}^*$.*

Since the map

$$\mathbf{M} \times \mathcal{S}_f \longrightarrow \chi + \mathfrak{m}^{\perp}$$

is an isomorphism of affine varieties,

$$\mathcal{S}_f \cong (\chi + \mathfrak{m}^{\perp})/\mathbf{M} = \mu^{-1}(\chi|_{\mathfrak{m}})/\mathbf{M}.$$

Therefore, the conditions of Theorem 3.27 are satisfied. We get Poisson structure on $\mathcal{S}_f$. It is described as follows.

Let $\pi\colon \chi + \mathfrak{m}^{\perp} \twoheadrightarrow (\chi + \mathfrak{m}^{\perp})/\mathbf{M} \simeq \mathcal{S}_f$ be the natural projection map, and $\iota\colon \chi + \mathfrak{m}^{\perp} \hookrightarrow \mathfrak{g}^*$ be the natural inclusion. Then for any $f, g \in \mathbb{C}[\mathcal{S}_f]$,

$$\{f, g\}_{\mathcal{S}_f} \circ \pi = \{\tilde{f}, \tilde{g}\} \circ \iota$$

where $\tilde{f}, \tilde{g}$ are arbitrary extensions of $f \circ \pi, g \circ \pi$ to $\mathfrak{g}^*$. From this, one can see that the latter Poisson structure defined on $\mathcal{S}_f$ is the same as that defined in 3.

Let

$$I_\chi = \mathbb{C}[\mathfrak{g}^*] \sum_{x \in \mathfrak{m}} (x - \chi(x))$$

where $\mathbb{C}[\mathfrak{g}^*]$ is identified with $\mathbb{C}[\mathfrak{g}]$, so that

$$\mathbb{C}[\mu^{-1}(\chi)] = \mathbb{C}[\mathfrak{g}^*]/I_\chi.$$

Then $\mathbb{C}[\mathcal{S}_f] = \mathbb{C}[\mu^{-1}(\chi)]^{\mathbf{M}}$ can be identified as the subspace of $\mathbb{C}[\mathfrak{g}^*]/I_\chi$ consisting of all cosets $\phi + I_\chi$ such that the infinitesimal action of M on ϕ $\{x, \phi\} \in I_\chi$ for all $x \in \mathfrak{m}$. In this realization, the Poisson structure on $\mathbb{C}[\mathcal{S}_f]$ is defined by the formula

$$\{\phi + I_\chi, \phi' + I_\chi\} = \{\phi, \phi'\} + I_\chi$$

for ϕ, ϕ' such that $\{x, \phi\}, \{x, \phi'\} \in I_\chi$ for all $x \in \mathfrak{m}$.

Example 3.31. Write a generic element of $\mathfrak{sl}_4$ as

$$X = (x_{ij})_{1\leq i,j\leq 4}, \qquad \sum_{i=1}^{4} x_{ii} = 0.$$

Because

$$\mathfrak{m} = \mathrm{span}\{E_{12}, E_{14}, E_{23}, E_{43}, E_{13}\},$$

and $\chi + \mathfrak{m}^{\perp} \in \mathfrak{g}^*$ is isomorphic to $f + \mathfrak{m}^{\perp}$.

So a generic point of $\chi + m^{\perp}$ has the form

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ 2 & x_{22} & x_{23} & x_{24} \\ 0 & 2 & x_{33} & 0 \\ 0 & x_{42} & x_{43} & x_{44} \end{pmatrix}, \qquad x_{44} = -(x_{11} + x_{22} + x_{33}).$$

Unfortunately I am not able to calculate the explicit expression of $\mathbb{C}[\mu^{-1}(\chi)]^{\mathbf{M}}$ in $\mathfrak{sl}_4$ case or any non-trivial cases. But we could still go on to state quantization in a abbreviate way.

3.4. **BRST Realization of Slodowy Slices and its Quantization.** [Ara17, §2.4]

Let $Cl(\mathfrak{g})$ be the *Clifford algebra* associated with the vector space $\mathfrak{g} \oplus \mathfrak{g}^*$ and the nondegenerate bilinear form $\langle \cdot | \cdot \rangle$ defined by

$$\langle f + x \mid g + y \rangle = f(y) + g(x)$$

for $f, g \in \mathfrak{g}^*$, $x, y \in \mathfrak{g}$. Namely, $Cl(\mathfrak{g})$ is the unital superalgebra isomorphic to

$$\wedge(\mathfrak{g} \oplus \mathfrak{g}^*) \cong \wedge(\mathfrak{g}) \otimes \wedge(\mathfrak{g}^*)$$

as $\mathbb{C}$-vector spaces, and the natural embeddings

$$\wedge(\mathfrak{g}) \hookrightarrow Cl(\mathfrak{g}), \qquad \wedge(\mathfrak{g}^*) \hookrightarrow Cl(\mathfrak{g})$$

are homogeneous homomorphisms of superalgebras.

Define an increasing filtration on $Cl(\mathfrak{g})$ by setting

$$Cl_p(\mathfrak{g}) := \wedge^{\leqslant p}(\mathfrak{g}) \otimes \wedge(\mathfrak{g}^*),$$

where

$$\wedge(\mathfrak{g}) = \bigoplus_{i\geqslant 0} \wedge^i(\mathfrak{g})$$

is the natural grading. We have

$$0 = Cl_{-1}(\mathfrak{g}) \subset Cl_0(\mathfrak{g}) \subset Cl_1(\mathfrak{g}) \subset \cdots \subset Cl_d(\mathfrak{g}) = Cl(\mathfrak{g}),$$

and

$$Cl_p(\mathfrak{g}) \cdot Cl_q(\mathfrak{g}) \subset Cl_{p+q}(\mathfrak{g}), \qquad [Cl_p(\mathfrak{g}), Cl_q(\mathfrak{g})] \subset Cl_{p+q-1}(\mathfrak{g}).$$

Let $\overline{Cl}(\mathfrak{g})$ be its associated graded algebra:

$$\overline{Cl}(\mathfrak{g}) := \operatorname{gr} Cl(\mathfrak{g}) = \bigoplus_{p \geqslant 0} \frac{Cl_p(\mathfrak{g})}{Cl_{p-1}(\mathfrak{g})}.$$

Then $\overline{Cl}(\mathfrak{g})$ is naturally a graded Poisson superalgebra, called the *classical Clifford algebra* associated with $\mathfrak{g} \oplus \mathfrak{g}^*$. Its Poisson superbracket is given by

$$\{x, f\} = f(x), \qquad x \in \mathfrak{g} \subset \wedge(\mathfrak{g}), \quad f \in \mathfrak{g}^* \subset \wedge(\mathfrak{g}^*),$$

$$\{x, y\} = 0, \qquad x, y \in \mathfrak{g} \subset \wedge(\mathfrak{g}),$$

$$\{f, g\} = 0, \qquad f, g \in \mathfrak{g}^* \subset \wedge(\mathfrak{g}^*).$$

Set

$$\bar{C}(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g}).$$

Then $\bar{C}(\mathfrak{g})$ is naturally a Poisson superalgebra, since it is a tensor product of Poisson superalgebras.

Lemma 3.32. *The following map gives a Lie algebra homomorphism:*

$$\begin{aligned} \bar{\theta}_\chi : \mathfrak{g} &\longrightarrow \bar{C}(\mathfrak{g}) \\ x &\longmapsto (\mu^*(x) - \chi(x)) \otimes 1 + 1 \otimes \bar{\rho}(x). \end{aligned}$$

Let

$$\bar{C}(\mathfrak{g}) = \bigoplus_{n \in \mathbb{Z}} \bar{C}^n(\mathfrak{g})$$

be the $\mathbb{Z}$-grading defined by

$$\deg(\phi \otimes 1) = 0 \quad (\phi \in \mathbb{C}[\mathfrak{g}^*]),$$

$$\deg(1 \otimes f) = 1 \quad (f \in \mathfrak{n}^*),$$

$$\deg(1 \otimes x) = -1 \quad (x \in \mathfrak{n}).$$

Then

$$\bar{C}^n(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \left(\bigoplus_{j-i=n} \Lambda^i(\mathfrak{g}) \otimes \Lambda^j(\mathfrak{g}^*) \right).$$

Lemma 3.33. *There exists a unique element $\bar{D} \in \bar{C}^1(\mathfrak{g})$ such that*

$$\{\bar{D}, 1 \otimes x\} = \bar{\theta}_\chi(x) \qquad \text{for } x \in \mathfrak{g}.$$

Moreover,

$$\{\bar{D}, \bar{D}\} = 0.$$

Remark 3.34. Actually,

$$\bar{D} = \sum_i (x_i - \chi(x_i)) \otimes x_i^* - 1 \otimes \frac{1}{2} \sum_{i,j,k} c_{i,j}^k \, x_i^* x_j^* x_k.$$

It is known that

$$\operatorname{ad} \bar{D} := \{\bar{D}, \cdot\}$$

satisfies

$$(\operatorname{ad} \bar{D})^2 = 0.$$

Thus,

$$(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

is a *differential graded Poisson superalgebra*. Its cohomology

$$H^\bullet_{BRST,\chi}(\mathfrak{g}, \mathbb{C}[\mathfrak{g}^*]) := H^\bullet(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D}) = \bigoplus_{i \in \mathbb{Z}} H^i(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

inherits a graded Poisson superalgebra structure from $\overline{C}(\mathfrak{g})$.

More generally, let P be a Poisson algebra equipped with a Poisson algebra homomorphism

$$\mu^* : \mathbb{C}[\mathfrak{g}^*] \to P.$$

View $\bar{D}$ as an element of

$$P \otimes \overline{Cl}(\mathfrak{g})$$

through the map

$$\mu^* \otimes \mathrm{id}.$$

Then

$$\mu^* \otimes \mathrm{id} : \overline{C}(\mathfrak{g}) \to P \otimes \overline{Cl}(\mathfrak{g})$$

is a Poisson superalgebra homomorphism, and

$$(P \otimes \overline{Cl}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

is a differential graded Poisson superalgebra, where the image of $\bar{D}$ is also denoted by $\bar{D}$. Therefore, its cohomology

$$H^\bullet_{BRST,\chi}(\mathfrak{g}, P) := H^\bullet(P \otimes \overline{Cl}(\mathfrak{g}), \operatorname{ad} \bar{D}) \tag{4}$$

inherits a graded Poisson superalgebra structure from

$$P \otimes \overline{Cl}(\mathfrak{g}).$$

Let $\mathfrak{m}$ be as in 1. We state without proof that

Theorem 3.35. *we have*

$$H^i_{BRST,\chi}(\mathfrak{m}, \mathbb{C}[\mathfrak{g}^*]) = 0 \qquad \textit{for } i \neq 0,$$

and

$$H^0_{BRST,\chi}(\mathfrak{m}, \mathbb{C}[\mathfrak{g}^*]) \cong \mathbb{C}[\mathcal{S}_f]$$

as Poisson algebras.

Let $\{U_i(\mathfrak{g})\}$ be the PBW filtration of the universal enveloping algebra $U(\mathfrak{g})$ of $\mathfrak{g}$. The PBW theorem gives an isomorphism of Poisson algebras:

$$\operatorname{gr} U(\mathfrak{g}) = \bigoplus_{i \geqslant 0} U_i(\mathfrak{g})/U_{i-1}(\mathfrak{g}) \cong \mathbb{C}[\mathfrak{g}^*].$$

Then we define

$$U(\mathfrak{g}, f) := H^0_{BRST,\chi}(\mathfrak{m}, U(\mathfrak{g})).$$

This is a quantization of $\mathbb{C}[\mathcal{S}_f]$. The algebra $U(\mathfrak{g}, f)$ is called the *finite W-algebra associated with f*.

W-algebra is itself a meaningful algebraic structure to study see example for [BK06; Lu+25; AE22]. And the more algebraic and well-known definition is via Whittaker Model.

3.5. **Whittaker Model Realization of Finite W-Algebras.** Let

$$\chi = (f \mid \cdot) \in \mathfrak{g}^*.$$

By restriction, χ defines a one-dimensional representation

$$\chi : U(\mathfrak{m}) \to \mathbb{C}.$$

We denote by $\mathbb{C}_\chi$ the corresponding left $U(\mathfrak{m})$-module.

Let I_χ be the left ideal of $U(\mathfrak{g})$ generated by the elements

$$x - \chi(x), \qquad x \in \mathfrak{m},$$

that is,

$$I_\chi := \sum_{x \in \mathfrak{m}} U(\mathfrak{g})(x - \chi(x)).$$

Then one has a natural isomorphism of left $U(\mathfrak{g})$-modules

$$Q_\chi := U(\mathfrak{g}) \otimes_{U(\mathfrak{m})} \mathbb{C}_\chi \cong U(\mathfrak{g})/I_\chi.$$

The $U(\mathfrak{g})$-module Q_χ is called the *generalized Gelfand–Graev module* associated with f.

Since the adjoint action of $\mathfrak{m}$ on $\mathfrak{g}$ extends to an action on $U(\mathfrak{g})$, and the ideal I_χ is $\mathfrak{m}$-stable, the quotient Q_χ carries a natural $\mathfrak{m}$-module structure. We define

$$Q_\chi^{\mathfrak{m}} := \{\bar{u} \in Q_\chi \mid [x, u] \in I_\chi \text{ for all } x \in \mathfrak{m}\},$$

where $\bar{u}$ denotes the image of $u \in U(\mathfrak{g})$ in Q_χ.

Definition 3.36. The finite W-algebra associated with f is defined by

$$U(\mathfrak{g}, f) := Q_\chi^{\mathfrak{m}}.$$

Equivalently,

$$U(\mathfrak{g}, f) = \{\bar{u} \in Q_\chi \mid [x, u] \in I_\chi \text{ for all } x \in \mathfrak{m}\}.$$

This realization of $U(\mathfrak{g}, f)$ is called the *Whittaker model realization.*

Remark 3.37. The space $Q_\chi^{\mathfrak{m}}$ is precisely the space of Whittaker vectors in Q_χ:

$$Q_\chi^{\mathfrak{m}} = \mathrm{Wh}(Q_\chi) := \{u \in Q_\chi \mid xu = \chi(x)u \text{ for all } x \in \mathfrak{m}\}.$$

We state the following theorem by Arakawa in [Ara07] without proof.

Theorem 3.38. *There are natural algebra isomorphisms*

$$U(\mathfrak{g}, f) \cong \mathrm{End}_{U(\mathfrak{g})}(Q_\chi)^{\mathrm{op}} \cong Q_\chi^{\mathfrak{m}}.$$

Here op *denotes the opposite algebra structure on* $\mathrm{End}_{U(\mathfrak{g})}(Q_\chi)$*, namely, the multiplication is given by reverse composition:*

$$\varphi \cdot \psi := \psi \circ \varphi.$$

4. Nakajima quiver variety

We mainly follow notations in [Kir16], [Maf00] and [New09].

4.1. **Quiver Representation and Quiver Variety.**

Definition 4.1 (The doubled quiver of type A_{n-1})**.** Fix an integer $n \geq 2$. The *doubled quiver of type* A_{n-1} is the quiver with vertex set

$$I = \{1, \ldots, n-1\},$$

and with arrows

$$i \xrightarrow{\alpha_i} i+1, \qquad i+1 \xrightarrow{\beta_i} i \qquad (1 \leq i \leq n-2).$$

Thus, for each edge of the Dynkin diagram of type A_{n-1}, we include one arrow in each direction.

$$\begin{tikzcd} 1 \arrow[r, bend left, "\alpha_1"] & 2 \arrow[l, bend left, "\beta_1"] \arrow[r, bend left, "\alpha_2"] & \cdots \arrow[l, bend left, "\beta_2"] \end{tikzcd} \qquad \begin{tikzcd} n-2 \arrow[r, bend left, "\alpha_{n-2}"] & n-1 \arrow[l, bend left, "\beta_{n-2}"] \end{tikzcd}$$

Example 4.2. For the Dynkin daigram of type A_3, we may begin with any orientation of the underlying graph

$$1 - 2 - 3.$$

For definiteness, choose the orientation

$$1 \longrightarrow 2 \longrightarrow 3.$$

We pass from this oriented quiver to the doubled quiver

$$1 \rightleftarrows 2 \rightleftarrows 3.$$

Definition 4.3 (Dimension vectors and framing data)**.** Let

$$d = (d_1, \ldots, d_{n-1}) \in \mathbb{Z}_{\geq 0}^{n-1}, \qquad v = (v_1, \ldots, v_{n-1}) \in \mathbb{Z}_{\geq 0}^{n-1}.$$

For each $i \in I$, let V_i and D_i be complex vector spaces such that

$$\dim V_i = v_i, \qquad \dim D_i = d_i.$$

The vector spaces D_i are called the *framing spaces.*

Definition 4.4 (The framed representation of double quivers)**.** With the notation above, define

$$S(d,v) := \bigoplus_{i=1}^{n-2} \mathrm{Hom}(V_i, V_{i+1}) \oplus \bigoplus_{i=1}^{n-2} \mathrm{Hom}(V_{i+1}, V_i) \oplus \bigoplus_{i=1}^{n-1} \mathrm{Hom}(D_i, V_i) \oplus \bigoplus_{i=1}^{n-1} \mathrm{Hom}(V_i, D_i).$$

An element of $S(d,v)$ will be written as

$$(A,B,\gamma,\delta) = \big((A_i)_{i=1}^{n-2}, (B_i)_{i=1}^{n-2}, (\gamma_i)_{i=1}^{n-1}, (\delta_i)_{i=1}^{n-1}\big),$$

where

$$A_i \in \mathrm{Hom}(V_i, V_{i+1}), \qquad B_i \in \mathrm{Hom}(V_{i+1}, V_i),$$
$$\gamma_i \in \mathrm{Hom}(D_i, V_i), \qquad \delta_i \in \mathrm{Hom}(V_i, D_i).$$

$$\begin{tikzcd}
D_1 \arrow[d, "\gamma_1"] & D_2 \arrow[d, "\gamma_2"] & \cdots & D_{n-2} \arrow[d, "\gamma_{n-2}"] & D_{n-1} \arrow[d, "\gamma_{n-1}"] \\
V_1 \arrow[r, bend left, "A_1"] \arrow[d, "\delta_1"] & V_2 \arrow[l, bend left, "B_1"] \arrow[r, bend left, "A_2"] \arrow[d, "\delta_2"] & \cdots \arrow[l, bend left, "B_2"] & V_{n-2} \arrow[r, bend left, "A_{n-2}"] \arrow[d, "\delta_{n-2}"] & V_{n-1} \arrow[l, bend left, "B_{n-2}"] \arrow[d, "\delta_{n-1}"] \\
D_1 & D_2 & & D_{n-2} & D_{n-1}
\end{tikzcd}$$

Next we consider the natural action of a group G_V. Set

$$G_V := \prod_{i=1}^{n-1} \mathrm{GL}(V_i).$$

The group G_V acts on $S(d,v)$ by

$$g \cdot (A,B,\gamma,\delta) = \big((g_{i+1}A_i g_i^{-1})_i,\ (g_i B_i g_{i+1}^{-1})_i,\ (g_i\gamma_i)_i,\ (\delta_i g_i^{-1})_i\big),$$

for $g = (g_i)_{i\in I} \in G_V$.

Besides, the vector space $S(d,v)$ is equipped with the symplectic form

$$\omega\big((A,B,\gamma,\delta),(A',B',\gamma',\delta')\big) := \sum_{i=1}^{n-2} \mathrm{Tr}(A_iB_i' - A_i'B_i) + \sum_{i=1}^{n-1} \mathrm{Tr}(\gamma_i\delta_i' - \gamma_i'\delta_i).$$

Definition 4.5 (Moment map and the ADHM equations)**.** With respect to the symplectic form ω, the action of G_V is Hamiltonian. Its moment map

$$\mu = \mu_{d,v} : S(d,v) \longrightarrow \bigoplus_{i=1}^{n-1} \mathfrak{gl}(V_i)$$

is given by

$$\mu(A,B,\gamma,\delta) = \big(\mu_1(A,B,\gamma,\delta), \ldots, \mu_{n-1}(A,B,\gamma,\delta)\big),$$

where

$$\begin{aligned}\mu_1 &= B_1A_1 - \gamma_1\delta_1,\\ \mu_i &= B_iA_i - A_{i-1}B_{i-1} - \gamma_i\delta_i \qquad (2 \le i \le n-2),\\ \mu_{n-1} &= -A_{n-2}B_{n-2} - \gamma_{n-1}\delta_{n-1}.\end{aligned}$$

Equivalently, on the zero-fibre $\mu^{-1}(0)$ the ADHM equations

$$\begin{aligned}B_1A_1 &= \gamma_1\delta_1,\\ B_iA_i &= A_{i-1}B_{i-1} + \gamma_i\delta_i \qquad (2 \le i \le n-2),\\ 0 &= A_{n-2}B_{n-2} + \gamma_{n-1}\delta_{n-1}\end{aligned}$$

are satisfied

We define

$$\Lambda(d,v) := \mu^{-1}(0) \subset S(d,v).$$

Definition 4.6 (Quiver variety)**.** The Quiver variety is defined by the affine quotient of $\Lambda(d,v)$ by G_V:

$$\mathfrak{M}_0(d,v) := \Lambda(d,v) /\!\!/ G_V := \mathrm{Spec}\big(\mathbb{C}[\Lambda(d,v)]^{G_V}\big).$$

Definition 4.7 (Characters of G_V)**.** For

$$\theta = (\theta_1, \dots, \theta_{n-1}) \in \mathbb{Z}^{n-1},$$

define the character

$$\chi_\theta : G_V \longrightarrow \mathbb{C}^\times$$

by

$$\chi_\theta(g) = \prod_{i=1}^{n-1} \det(g_i)^{-\theta_i}.$$

Definition 4.8 (Nakajima quiver variety)**.** The quiver variety with parameter θ is defined by

$$\mathfrak{M}_\theta(d,v) := \Lambda(d,v) /\!\!/_{\chi_\theta} G_V := \mathrm{Proj}\big(R_\theta(d,v)\big).$$

Example 4.9. By 2.16 we could find the explicit expression for above definition.

For each integer $m \ge 0$, let

$$\mathbb{C}[\Lambda(d,v)]^{G_V,\chi_\theta^m} := \{f \in \mathbb{C}[\Lambda(d,v)] \mid f(g\cdot x) = \chi_\theta(g)^m f(x) \text{ for all } g \in G_V,\ x \in \Lambda(d,v)\}.$$

We then obtain the graded ring of χ_θ-semi-invariants

$$R_\theta(d,v) := \bigoplus_{m\geq 0} \mathbb{C}[\Lambda(d,v)]^{G_V,\chi_\theta^m}.$$

Definition 4.10. Let

$$\mathbf{1} := (1,\dots,1) \in \mathbb{Z}^{n-1}.$$

We define

$$\mathfrak{M}(d,v) := \mathfrak{M}_{\mathbf{1}}(d,v).$$

The inclusion

$$\mathbb{C}[\Lambda(d,v)]^{G_V} \hookrightarrow R_{\mathbf{1}}(d,v)$$

induces a natural projective morphism

$$\pi : \mathfrak{M}(d,v) \longrightarrow \mathfrak{M}_0(d,v).$$

We denote by

$$\mathfrak{M}_1(d,v) := \mathrm{Im}(\pi)$$

the image of π, endowed with its reduced induced structure.

4.2. **Nakajima's Conjecture.**

Definition 4.11 ($\mathfrak{sl}_2$-triple and Slodowy slice)**.** Let $x \in \mathcal{N}$. An $\mathfrak{sl}_2$-*triple* containing x is a triple

$$(x,y,h) \in \mathfrak{gl}(D)^3$$

satisfying

$$[h,x] = 2x, \qquad [h,y] = -2y, \qquad [x,y] = h.$$

Fix such a triple. Denote the *Slodowy slice* to $\mathcal{N}$ at x by $\mathcal{S}_x$

Definition 4.12 (Partial flag variety of type a)**.** Let

$$a = (a_1,\dots,a_n) \in \mathbb{Z}^n_{\geq 0} \qquad \text{with} \qquad \sum_{i=1}^n a_i = N.$$

A *partial flag of type* a in D is a chain of subspaces

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = D$$

such that

$$\dim(F_i/F_{i-1}) = a_i \qquad (1 \leq i \leq n).$$

The corresponding partial flag variety is denoted by

$$\mathcal{F}_a.$$

Definition 4.13 (Cotangent bundle realization and Springer-type map)**.** Let $a = (a_1, \dots, a_n) \in \mathbb{Z}^n_{\geq 0}$ with $\sum_i a_i = N$. Define

$$\widetilde{\mathcal{N}}_a := T^*\mathcal{F}_a.$$

It admits the standard realization

$$\widetilde{\mathcal{N}}_a \cong \big\{(u, F) \in \mathfrak{gl}(D) \times \mathcal{F}_a \;\big|\; u(F_i) \subset F_{i-1} \text{ for all } i\big\}.$$

Define the morphism

$$\mu_a : \widetilde{\mathcal{N}}_a \longrightarrow \mathcal{N}, \qquad (u, F) \longmapsto u.$$

Definition 4.14 (The partition λ_a and the varieties $S_{a,x}$, $\widetilde{S}_{a,x}$)**.** Let

$$a = (a_1, \dots, a_n) \in \mathbb{Z}^n_{\geq 0}, \qquad \sum_{i=1}^n a_i = N.$$

Let

$$\alpha = (\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n)$$

be the nonincreasing rearrangement of a. Define the partition

$$\lambda_a = 1^{\alpha_1 - \alpha_2} 2^{\alpha_2 - \alpha_3} \cdots n^{\alpha_n}.$$

Let

$$\mathcal{O}_{\lambda_a} \subset \mathcal{N}$$

be the nilpotent orbit corresponding to the partition λ_a, and let

$$\overline{\mathcal{O}_{\lambda_a}}$$

be its Zariski closure. For a fixed nilpotent element $x \in \mathcal{N}$, define

$$S_{a,x} := S_x \cap \overline{\mathcal{O}_{\lambda_a}}, \qquad \widetilde{S}_{a,x} := \mu_a^{-1}(S_{a,x}).$$

Conjecture 4.15 (Nakajima)**.** *Let*

$$d = (d_1, \dots, d_{n-1}) \in \mathbb{Z}^{n-1}_{\geq 0}, \qquad v = (v_1, \dots, v_{n-1}) \in \mathbb{Z}^{n-1},$$

and define

$$N = \sum_{i=1}^{n-1} i\, d_i, \qquad a = a(d, v) \in \mathbb{Z}^n$$

as above. Let D be a complex vector space of dimension N, and let

$$x \in \mathcal{N} \subset \mathfrak{gl}(D)$$

be a nilpotent element of Jordan type

$$1^{d_1} 2^{d_2} \cdots (n-1)^{d_{n-1}}.$$

Then there exist isomorphisms of algebraic varieties

$$\widetilde{\varphi} : \mathfrak{M}(d,v) \xrightarrow{\sim} \widetilde{S}_{a,x}, \qquad \varphi_1 : \mathfrak{M}_1(d,v) \xrightarrow{\sim} S_{a,x},$$

such that the point

$$0 \in \mathfrak{M}_1(d,v)$$

is mapped to

$$x \in S_{a,x},$$

and such that the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{M}(d,v) & \xrightarrow{\widetilde{\varphi}} & \widetilde{S}_{a,x} \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle \mu_a} \\ \mathfrak{M}_1(d,v) & \xrightarrow[\varphi_1]{} & S_{a,x}. \end{array}$$

5. Affine Grassmannian

Let G be a complex semisimple group and consider the affine Grassmannian $\mathrm{Gr}_G = G((t^{-1}))/G[t]$ of G. Inside the affine Grassmannian, we have the spherical Schubert cells

$$\mathrm{Gr}^\lambda := G[t]t^\lambda,$$

for λ a dominant coweight of G. We also have transverse orbits

$$\mathcal{W}_\mu := G_1[[t^{-1}]]t^\mu,$$

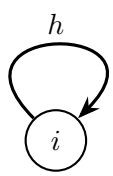
for μ a dominant coweight, and we can study their intersections

$$\overline{\mathcal{W}}_\mu^\lambda := \mathcal{W}_\mu \cap \overline{\mathrm{Gr}^\lambda},$$

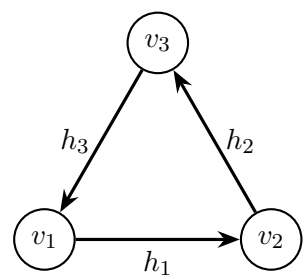
which are known as *affine Grassmannian slices.*

6. Split Quotient

Let(I, H) be a finite graph without edge loops. Here I denotes the set of vertices and H denotes the set of edges together with their orientations, with $s, t : H \to I$ being the source and target functions and $^{-} : H \to H$ the involution given by reversing orientation. Let Ω be an orientation of (I, H), i.e. a subset of H such that H is the disjoint union of Ω and $\overline{\Omega}$, and assume that Ω contains no oriented cycle. The triple (I, H, Ω) is what we refer to as a quiver.



Edge Loop
(Forbidden)
$s(h) = t(h)$



Oriented Cycle
(Forbidden in Ω)
$v_1 \to v_2 \to v_3 \to v_1$

Following Lusztig, we consider an *admissible automorphism* a of (I, H). This comprises permutations of I and H, each denoted a, such that $s(a(h)) = a(s(h))$, $t(a(h)) = a(t(h)), \overline{a(h)} = a(\overline{h})$, and no two adjacent vertices belong to the same $\langle a \rangle$-orbit. We assume moreover that $a(\Omega) = \Omega$; given (I, H, a), there is always at least one orientation Ω compatible with a in this sense.

For any $i \in I$, let d_i be the smallest positive integer such that $a^{d_i(i)} = i$. For $h \in H$, define d_h similarly; clearly d_h is a common multiple of $d_{s(h)}$ and $d_{t(h)}$ (we do not assume that it is the least common multiple, i.e. we allow two edges in the same $\langle a \rangle$-orbit to have the same source and target). Let $\mathbf{n}$ be a common multiple of all d_i and d_h. Set $e_i = \mathbf{n}/d_i$, $e_h = \mathbf{n}/d_h$.

We define a quiver $(\widetilde{I}, \widetilde{H}, \widetilde{\Omega})$,which we call the $split - quotient$ quiver of $(I, H, \Omega, a, \mathbf{n})$. First we form the quotient quiver $(\widehat{I}, \widehat{H}, \widehat{\Omega})$ of (I, H, Ω) by $\langle a \rangle$. Namely, let $\widehat{I}$ denote a set of representatives for the $\langle a \rangle$-orbits in I,let $\widehat{H}$ be the set of $\langle a \rangle$-orbits in H,and let $\widehat{\Omega} \subset \widehat{H}$ be the set of $\langle a \rangle$-orbits in

Ω. (The reason we have defined $\widehat{I}$ to be a set of representatives for the orbits, rather than the set of orbits itself, is just notational convenience.) For $\widehat{h} = \langle a\rangle \cdot h \in \widehat{H}$, define $s(\widehat{h})$ to be the representative in notational convenience.) For $\widehat{h} = \langle a\rangle \cdot h \in \widehat{H}$, define $s(\widehat{h})$ to be the representative in $\widehat{I}$ of $\langle a\rangle \cdot s(h)$, defne $t(\widehat{h})$ similarly, and define $d_{\widehat{h}} = d_h, e_{\widehat{h}} = e_h$. Define$^- : \widehat{H} \to \widehat{H}$ by $\overline{\langle a\rangle \cdot h} = \langle a\rangle \cdot \overline{h}$. Then $(\widehat{I}, \widehat{H}, \widehat{\Omega})$ is another quiver with no edge-loops or directed cycles.

Now let $\widetilde{I}$ be the set of pairs (i, ζ) where $i \in \widehat{I}$ and $\zeta \in \mu_{e_i}$ (the group of complex (e_i)th roots of 1). Let $\widetilde{H}$ be the set of triples $(\widehat{h}, \zeta, \zeta')$ where $\widehat{h} \in \widehat{H}, \zeta \in \mu_{e_{s(\widehat{h})}}$, $\zeta' \in \mu_{e_{t(\widehat{h})}}$, and

$$\zeta^{e_{s(\widehat{h})}/e_{\widehat{h}}} \;=\; (\zeta')^{e_{t(\widehat{h})}/e_{\widehat{h}}}.$$

Let $\widetilde{\Omega}$ be the set of $(\widehat{h}, \zeta, \zeta') \in \widetilde{H}$ such that $\widehat{h} \in \widehat{\Omega}$. For $\widetilde{h} = (\widehat{h}, \zeta, \zeta') \in \widetilde{H}$, define $s(\widetilde{h}) = (s(\widehat{h}), \zeta)$ and $t(\widetilde{h}) = (t(\widehat{h}), \zeta')$. Define$- : \widetilde{H} \to \widetilde{H}$ by $\overline{(\widehat{h}, \zeta, \zeta')} = (\widehat{\overline{h}}, \zeta', \zeta)$. Then $(\widetilde{I}, \widetilde{H}, \widetilde{\Omega})$ is another quiver with no edge-loops or oriented cycles.

Example 6.1. Fix a positive integer n. Let (I, H, Ω) be the quiver of type $\mathrm{A}_2 n - 1$ defined as follows: $I = \{1, 2, \cdots, 2n-1\}$, there is an edge in H from i to j if and only if $|i - j| = 1$, and Ω consists of the oriented edges h such that $t(h)$ is closer to the middle vertex n than $s(h)$ is. Let a be the involution of the quiver that interchanges i and $2n - i$. Then d_i equals 2 for $i \neq n$ and 1 for $i = n$, and $d_h = 2$ for all h. We take $\mathbf{n} = 2$, so e_i equals 1 for $i \neq n$ and 2 for $i = n$, and $e_h = 1$ for all h.

The quotient quiver $(\widehat{I}, \widehat{H}, \widehat{\Omega})$ is then of type A_n: we set $\widehat{I} = \{1, \cdots, n\}$, so again there is an edge in $\widehat{H}$ from i to j if and only if $|i - j| = 1$, and $\widehat{\Omega}$ consists of the edges from i to $i+1$ for $1 \leq i \leq n-1$. Since $e_i = 1$ for all $i \neq n$, the only modification required to obtain the split-quotient quiver $(\widetilde{I}, \widetilde{H}, \widetilde{\Omega})$ is that the vertex n should be split into two vertices $(n, +1)$ and $(n, -1)$ respectively and related edges are $(\widetilde{h}, 1, +1)$ and $(\widetilde{h}, 1, -1)$. Thus $(\widetilde{I}, \widetilde{H}, \widetilde{\Omega})$ is a quiver of type D_{n+1}. We show in the following figure the behavior of Ω:

$$(I, H, \Omega): \; 1 \longrightarrow \cdots\cdots \rightarrow n-1 \longrightarrow n \longleftarrow n+1 \leftarrow \cdots\cdots \leftarrow 2n-1$$

$$
\begin{tikzcd}
(\widehat{I},\widehat{H},\widehat{\Omega}): & \underset{\{1,2n-1\}}{1} \arrow[r] & \cdots\cdots \arrow[r] & \underset{\{n-1,n+1\}}{n-1} \arrow[r] & \underset{\{n\}}{n} \\
(\widetilde{I},\widetilde{H},\widetilde{\Omega}): & 1 \arrow[r] & \cdots\cdots \arrow[r] & n-1 \arrow[ur] \arrow[dr] & + \arrow[d, dotted, bend left] \\
 & & & & -
\end{tikzcd}
$$

7. Beyond Type A

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